Abstract Algebra Notes

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0 Introduction

This document is a compilation of all notes taken in Dr. Craig Timmons' graduate level Abstract Algebra Class at California State University, Sacramento. Many of the sections follow those of Dummit and Foote's Abstract Algebra, 3rd Edition, though the notes are not directly aligned with the book. Specifically, many sections are ommitted, and many definitions, theorems, and examples are not from the book and do not follow a numbering system that is consistent with the book. Still, these notes may be very useful to students who are taking graduate level Abstract Algebra coursework.

For my comprehensive exams. Good luck G.

1 Chapter 1: Introduction to Groups

1.1 Basic Axioms and Examples

Definition 1.1.1: A group is a nonempty set G with a binary operation $*$ from $G \times G \longrightarrow G$, where $G \times G = \{(g, h) : g \in G, h \in H\}$. We may represent $*$ using notation that's more algebraic. For instance, we may write $g \times h$ or gh instead of $f(g, h)$.

Example 1.1.1: A binary operation on $S = \{1, 2, 3\}$ could be

$*$	1	2	3
1	1	1	1
2	1	1	1
3	1	1	1

A set G with a binary operation $*$ is a group if

1. $a * (b * c) = (a * b) * c \ \forall a, b, c \in G$
2. $\exists e \in G$ such that $a * e = a = e * a$
3. for each $a \in G$, $\exists b \in G$ such that $a * b = e = b * a$.

That is, G is a group if G is associative, contains an identity, and contains inverses for each element of G. We say G is abelian if $a * b = b * a \ \forall a, b \in G$ (G is commutative). We say G is a finite group if G as a set is finite (otherwise, G is infinite).

Example 1.1.2: Some examples of groups include:

1. $(\mathbb{Z}, +)$: integers under addition,
2. $(\mathbb{Q}, +)$: rationals under addition,
3. $(\mathbb{R}, +)$: reals under addition,
4. $(\mathbb{C}, +)$: complex under addition,

5. $(\mathbb{M}_{2\times 2}\mathbb{R}, +)$: real 2 by 2 matrices under addition,
6. $(\mathbb{Z}, \cdot)$: integers under multiplication,
7. $(\mathbb{Q}\{0\}, \cdot)$: rationals (not including 0) under multiplication,
8. $(\mathbb{R}\{0\}, \cdot) = \mathbb{R}^*$: reals (not including 0) under multiplication, and
9. for $n \in \mathbb{Z}^+$, $\mathbb{Z}/n\mathbb{Z} = \{\text{equivalence classes under the relation } a \sim b \longleftrightarrow n|(a-b), a, b \in \mathbb{Z}\}$.

Definition 1.1.2: The equivalence class containing a is denoted $\overline{a}$. We have $a = \{a + nk : k \in \mathbb{Z}\}$. Note that the operation is $\overline{a} + \overline{b} = \overline{a+b}$, where $\overline{a} + \overline{b}$ is addition under the group operation while $\overline{a+b}$ is the equivalence class containing $a + b$ where $a + b$ is the usual addition of integers.

Example 1.1.3:

$\mathbb{Z}_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

The example above is the group of integers modulo n. Note that in this case, $n = 4$. For $n \in \mathbb{Z}^+$, $\left(\mathbb{Z}/n\mathbb{Z}\right)^* = \{\overline{a} \in \mathbb{Z}/n\mathbb{Z} : \gcd(a, n) = 1\}$ and the operation is $\overline{a} \cdot \overline{b} = \overline{a \cdot b}$.

Example 1.1.4: Some examples of modular groups include

1. $\mathbb{Z}^* = (\{-1, 1\}, \cdot)$,
2. $\mathbb{Z}_2 = (\{\overline{1}\}, \cdot)$,
3. $\mathbb{Z}_3 = (\{\overline{1}, \overline{2}\}, \cdot)$,
4. $\mathbb{Z}_4 = (\{\overline{1}, \overline{3}\}, \cdot)$, and
5. $GL_2(\mathbb{R}) = M_{2\times 2}(\mathbb{R})^*$.

Note that $GL_2(\mathbb{R})$ is the group of all invertible 2×2 matrices with real enteries under matrix multiplication and $M_{2\times 2}(\mathbb{R})^*$ is the group of all 2×2 matrices with real entries under addition.

Definition 1.1.3: A direct product of two groups $(A, *)$ and $(B, \circ)$ is the group whose elements are (a, b) where $a \in A,\ b \in B$, and the operation is $(a, b)(c, d) = (a * c, b \circ d)$.

Proposition 1.1.1: If $(G, \cdot)$ is a group, then

1. e is unique in G,
2. for each $a \in G$, a^{-1} is unique,
3. $(a^{-1})^{-1} = a\ \forall a \in G$,
4. $(ab)^{-1} = b^{-1}a^{-1}\ \forall a, b \in G$, and
5. for any $a_1, a_2, \ldots, a_n \in G$, their product $a_1 \cdot a_2 \cdots a_n$ does not depend on how the expression is bracketed.

Now, some comments on notation in groups. First, note that the default setting for group operations is multiplicative. That is, if G is any group with $x \in G$, then

$$\begin{aligned} x^0 &= e, \\ x^n &= x \cdots x \text{ } n \text{ times}, n \in \mathbb{Z}^+, \text{ and} \\ x^{-n} &= x^{-1} \cdot x^{-1} \cdots x^{-1} \text{ } n \text{ times}, n \in \mathbb{Z}^+ \\ &= (x \cdot x \cdots x)^{-1} \text{ with } n \text{ many } x\text{'s}. \end{aligned}$$

Next, note that if G is abelian, we may change to additive notation. That is,

$$\begin{aligned} a^{-1} &= -a, \\ ab &= a+b, \\ a^n &= a+a+\cdots+a \ n \text{ times}, \ n \in \mathbb{Z}^+, \\ e &= 0, \text{ and} \\ a^{-n} &= -na \\ &= n(-a) \\ &= (-a)+(-a)+\cdots+(-a) \ n \text{ times}, \ n \in \mathbb{Z}^+. \end{aligned}$$

Definition 1.1.4: Let D be a nonempty set. The power set of D, $\mathcal{P}(D)$, is the set of all possible subsets of D.

Example 1.1.5: Let $D = \{a, b, c\}$. Then

$$\mathcal{P}(D) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, D\}.$$

Example 1.1.6: Define $A + B = (A \backslash B) \cup (B \backslash A)$ as an additive operation. Then

$$\begin{aligned} A + \emptyset &= (A \backslash \emptyset) \cup (\emptyset \backslash A), \\ A^{-1} &= A \\ &\implies A + A = 2A = \emptyset. \end{aligned}$$

Then it turns out that $(\mathcal{P}(D), +)$ is an abelian group with $e^+ = \emptyset$. Observe that for any $A \in \mathcal{P}(D)$,

$$\begin{aligned} A + A &= (A \backslash A) \cup (A \backslash A) \\ &= \emptyset \\ &= 2A. \end{aligned}$$

Note that every element then has order 2 or 1 (we will discuss order in the near future).

Example 1.1.7: Let G be a finite group. The operation table for $G = \{g_1, g_2, \ldots, g_n\}$ is

$\cdot$	g_1	g_2	$\cdots$	g_j	$\cdots$	g_n
g_1	$g_1 \cdot g_1$	$g_1 \cdot g_2$	$\cdots$	$g_1 \cdot g_j$	$\cdots$	$g_1 \cdot g_n$
g_2	$g_2 \cdot g_1$	$g_2 \cdot g_2$	$\cdots$	$g_2 \cdot g_j$	$\cdots$	$g_2 \cdot g_n$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\cdots$	$\cdots$	$\vdots$
g_i	$g_i \cdot g_1$	$g_i \cdot g_2$	$\cdots$	$g_i \cdot g_j$	$\cdots$	$g_i \cdot g_n$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\cdots$	$\cdots$	$\vdots$
g_n	$g_i \cdot g_1$	$g_n \cdot g_2$	$\cdots$	$g_n \cdot g_j$	$\cdots$	$g_n \cdot g_n$

Definition 1.1.5: If G is a group with $x \in G$, then the order of x, denoted $|x|$, is the smallest positive integer n such that $x^n = e$. If no such n exists, then x has infinite order.

Example 1.1.8: Let $\mathbb{C}^*$ be the group of nonzero complex numbers under multiplication. This group has elements of infinite order, such as 2 or $2 + i$. There is also an element of order 1 — namely e^*. There is also an element of order 2 (-1), and an element of order 4 (i).

1.2 Dihedral Groups and Group Actions

Definition 1.2.1: Let G be a group and A a set. A group action of G on A is a function from $G \times A \longrightarrow A$ denoted $g \cdot a$ that satisfies

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a \ \forall g_1, g_2 \in G$ and $\forall a \in A$ and

2. $e \cdot a = a \ \forall a \in A$.

Note that for the items above, $\cdot$ denotes the group action, while the absence of $\cdot$ denotes the group operation on G. Further, e is the identity in G.

Example 1.2.1: Take $\mathbb{R}^*$ and $\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n)\, x_i \in \mathbb{R}\}$. Then

$$\alpha(x_1, x_2, \ldots, x_n) = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)$$

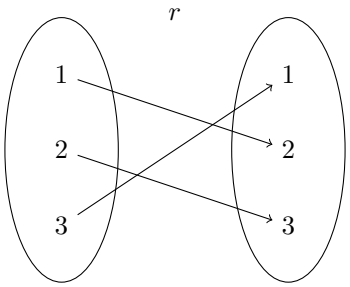
is a group action in which the group operation is scalar multiplication.

Definition 1.2.2: For $n \geq 3$, the Dihedral Group of order $2n$, denoted D_{2n}, is the set of symmetries of a regular n-gon.

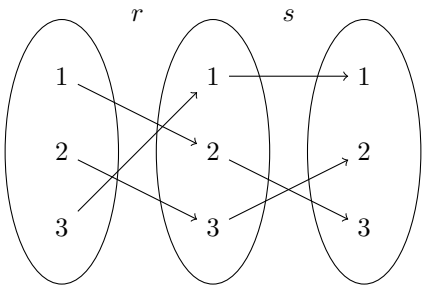
Example 1.2.2: $D_6 = D_{2\cdot 3}$ is the set of symmetries of a regular 3-gon (that is, of an equilateral triangle). The 6 symmetries are

1. rotate clockwise by 120°,
2. rotate clockwise by 240°,
3. reflect about vertical line of symmetry,
4. reflect about left corner,
5. reflect about right corner, and
6. keep it stationary.

Let r correspond to 120° rotation and let s correspond to the verticle reflection. Then we have $D_6 = \{r, r^2, e, s, sr, sr^2\}$. Think of r and s as functions. That is,



Thus, we treat the dihedral group of order 6 (or of any other order) like function composition. That is, $sr = s \circ r$, so



The notation we will use with D_{2n} is $D_{2n} = \{e, r, r^2, \ldots, r^{n-1}, s, sr, \ldots, sr^{n-1}\}$.

Definition 1.2.3: Let G be a group acting on a set A with $a \in A$. The stabilizer is the set $Stab(a) = \{g \in G : g \cdot a = a\}$. Next, we prove that the stabilizer forms a subgroup.

Proof: Let G and A be defined as above with $a \in A$. Since G is a group, $e \in G$. Further, $e \cdot a = a$, so $e \in Stab(a)$, and thusly $Stab(a) \neq \emptyset$.

Next, let $g, h \in Stab(a)$. Then $g \cdot a = a$ and $h \cdot a = a$. We want to show $(gh) \cdot a = a$. By definition of a group action, we know

$$\begin{aligned}(gh) \cdot a &= g \cdot (h \cdot a) \\ &= g \cdot a \\ &= a\end{aligned}$$

Then $(gh) \cdot a = a$, so $(gh) \in Stab(a)$. Next, we need to show $g^{-1} \in Stab(a)$. Consider $g \cdot a = a$. Then we have

$$\begin{aligned}g^{-1} \cdot a &= g^{-1} \cdot (g \cdot a) \\ &= (g^{-1}g) \cdot a \\ &= e \cdot a \\ &= a\end{aligned}$$

Then, since $a \in Stab(a)$, we must have that $g^{-1} \in Stab(a)\ \forall g \in Stab(a)$. Then $Stab(a) \leq G$. $\square$

Example 1.2.4: Let G act on G by the rule $g \cdot x = gxg^{-1}$. Let's check that this defines an action. For any $x \in G$, $e \cdot x = e \cdot x \cdot e^{-1} = x$. Next, for any $g, h \in G$ and $x \in G$, we have

$$\begin{aligned}(g \cdot h) \cdot x &= g \cdot (h \cdot x) \\ &= g \cdot (hxh^{-1}) \\ &= g(hxh^{-1})g^{-1} \\ &= (gh)x(h^{-1}g^{-1}) \\ &= (gh)x(gh)^{-1} \\ &= (gh)x\end{aligned}$$

so we must have that $g \cdot x = gxg^{-1}$ defines an action of G on G. Next, what is $Stab(x)$?

$$\begin{aligned}Stab(x) &= \{g \in G : g \cdot x = x\} \\ &= \{g \in G : gxg^{-1} = x\} \\ &= \{g \in G : gx = xg\} \\ &= C(x) \text{ or } C_G(x)\end{aligned}$$

Definition 1.2.4: Let G be a group with $g \in G$ and $x \in A$ with A a set. The collection of elements in G that commute with x is called the centralizer of x and is denoted $C(x)$ or $C_G(x)$. Similarly, if S is a set, then $C(S) = \{g \in G : gx = xg\ \forall x \in S\}$. Note that this is not to be confused with the center of G, defined below.

Definition 1.2.5: Let G be a group. Then the center of G is $Z(G) = \{g \in G : gx = xg\ \forall x \in G\}$. That is, the center of G is the set of all elements in G which commute with every element of G. Note that $Z(G) = G \implies G$ is abelian.

Now, some notes on notation and a relation that is used in D_{2n}. Remember, $D_{2n} = \{e, r, r^2, \ldots, r^{n-1}, s, sr, sr^2, \ldots, sr^{n-1}\}$. Then $|r| = n$ and $|s| = 2$, where r denotes a rotation and s denotes a reflection. Further, $r^i s = sr^{-i}$ $\forall 0 \leq i \leq n$. In particular, when $i = 1$, $rs = sr^{-1} = sr^{n-1}$. Similarly, $r^{n-1}s = sr$.

Example 1.2.5: Let $n = 5$. Then $r^3 s = sr^{-3} = sr^2$ with $D_{2\cdot 5} = \{e, r, r^2, r^3, r^4, s, sr, sr^2, sr^3, sr^4\}$. Next, let $n = 15$. Then $rsr^3 = rr^{12}s = r^{13}s$. Lastly, let $n = 12$. Then

$$\begin{aligned}(sr^9)(sr^6) &= sr^9r^{-6}s \\ &= sr^3s \\ &= r^{-3}ss \\ &= r^9e \\ &= r^9.\end{aligned}$$

1.3 Symmetric Groups

Definition 1.3.1: Let Ω be a nonempty set. Let S_Ω be the set of all bijections from Ω to Ω. The set S_Ω forms a group, the symmetric group on Ω, under function composition. The identity function in this set is denoted $\mathbb{1}$, or sometimes $Id(S_\Omega)$. What this means is $\mathbb{1}(x) = x \ \forall x \in \Omega$.

In a special case that this set $\Omega = \{1, 2, 3, \ldots, n\}$, we write S_n instead of $S_{1,2,3,\ldots,n}$. S_n is called the symmetric group of degree n. Elements in S_n are called permutations.

Example 1.3.1: One way we can represent permutations is with two line notation, as seen below:

Let $\alpha \in S_7$ such that $\alpha(1) = 5, \alpha(2) = 1, \alpha(3) = 7, \alpha(4) = 6, \alpha(5) = 2, \alpha(6) = 4$, and $\alpha(7) = 3$. Then the two line notation for α looks like

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 1 & 7 & 6 & 2 & 4 & 3 \end{pmatrix}.$$

Example 1.3.2: In S_6, if $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 2 & 4 & 3 & 6 \end{pmatrix}$, then $\beta = (2\ 5\ 3)$.

Definition 1.3.2: Two cycles $(a_1\ a_2\ \cdots\ a_n)$ and $(b_1\ b_2\ \cdots b_m)$ are disjoint if $\{a_1, a_2, \ldots, a_n\} \cap \{b_1, b_2, \ldots, b_m\} = \emptyset$.

Example 1.3.3: In S_7, $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 1 & 7 & 6 & 2 & 4 & 3 \end{pmatrix} = (1\ 5\ 2)(3\ 7)(4\ 6)$ and in S_{13},

$$\begin{aligned} \theta &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ 12 & 13 & 3 & 1 & 11 & 9 & 5 & 10 & 6 & 4 & 7 & 8 & 2 \end{pmatrix} \\ &= (1\ 12\ 8\ 10\ 4)(2\ 13)(3)(5\ 11\ 7)(6\ 9) \\ &= (1\ 12\ 8\ 10\ 4)(2\ 13)(5\ 11\ 7)(6\ 9). \end{aligned}$$

It is useful to note that 1-cycles are typically ommitted from products of cycles. Further, it is important to note that disjoint cycles commute.

Definition 1.3.3: The length of cycle $(a_1\ a_2\ \cdots\ a_n)$ is n, and this is commonly referred to as an n-cycle. Further, note that 2-cycles are also called transpositions.

Example 1.3.4: Let us write all elements of s_4 as products of disjoint cycles. First, we start with the 1-cycle — namely, $\{(1)\}$. Note that we could have written this as $\{(2)\}$, $\{(3)\}$, or $\{(4)\}$. Next, we consider the 2-cycles. We have

$$\{(1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4)\}.$$

The set of 3-cycles includes

$$\{(1\ 2\ 3), (1\ 3\ 2), (1\ 2\ 4), (1\ 4\ 2), (1\ 3\ 4), (1\ 4\ 3), (2\ 3\ 4), (2\ 4\ 3)\}.$$

Next, the set of 4-cycles is

$$\{(1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 3\ 2\ 4), (1\ 3\ 4\ 2), (1\ 4\ 2\ 3), (1\ 4\ 3\ 2)\}.$$

The last cycle-type present in S_4 is the 2-2-cycle, of which there are three: $\{(1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$. Note that there is one 1-cycle and there are six 2-cycles, eight 3-cycles, and six 4-cycles. This brings us to an interesting question — how many permutations in S_n have a given cycle type?

Example 1.3.5: How many permutations have each cycle type in S_5?

Cycle Type	Number of permutations with given cycle type	example
1-cycle	1	(1)
2-cycle	$\frac{5\cdot 4}{2} = 10$	$(1\ 2)$
3-cycle	$\frac{5\cdot 4\cdot 3}{3} = 20$	$(1\ 2\ 3)$
4-cycle	$\frac{5\cdot 4\cdot 3\cdot 2}{4} = 30$	$(1\ 2\ 3\ 4)$
5-cycle	$\frac{5\cdot 4\cdot 3\cdot 2\cdot 1}{5} = 24$	$(1\ 2\ 3\ 4\ 5)$
3-2-cycle	$\frac{5\cdot 4\cdot 3}{3}\cdot\frac{2\cdot 1}{2} = 20$	$(1\ 2\ 3)(4\ 5)$
2-2-cycle	$\frac{\frac{5\cdot 4}{2}\cdot\frac{3\cdot 2}{2}}{2} = 15$	$(1\ 2)(3\ 4)$

Note that the total number of permutations in S_n is $n!$.

Example 1.3.6: Consider S_{50}. How many 2-3-3-cycles are there?

$$\frac{50\cdot 49}{2}\cdot\frac{\frac{48\cdot 47\cdot 46}{3}\cdot\frac{45\cdot 44\cdot 43}{3}}{2} = 601,304,088,000.$$

Definition 1.3.3: Let G be a group. For $x, y \in G$, lets say that x is a conjugate of y if $x = gyg^{-1}$ for some $g \in G$. Another notation for this is to say that $x \in cl(y)$, or $x \sim y$. Further, the conjugacy class that contains y is $cl(y) = \{gyg^{-1} : g \in G\}$.

Example 1.3.7: In S_3, what is $cl((1\ 2))$? To find out, we must find all unique conjugates of $(1\ 2)$. We have

$$\begin{aligned}
(1)(1\ 2)(1)^{-1} &= (1\ 2)\\
(1\ 2)(1\ 2)(1\ 2)^{-1} &= (1\ 2)\\
(1\ 3)(1\ 2)(1\ 3)^{-1} &= (1\ 3)(1\ 2)(3\ 1) = (1)(2\ 3) = (2\ 3)\\
(2\ 3)(1\ 2)(2\ 3)^{-1} &= (2\ 3)(1\ 2)(3\ 2) = (1\ 3)(2) = (1\ 3)\\
(1\ 2\ 3)(1\ 2)(1\ 2\ 3)^{-1} &= (1\ 2\ 3)(1\ 2)(3\ 2\ 1) = (1)(2\ 3) = (2\ 3)\\
(1\ 3\ 2)(1\ 2)(1\ 3\ 2)^{-1} &= (1\ 3\ 2)(1\ 2)(2\ 3\ 1)^{-1} = (1\ 3)(2) = (1\ 3).
\end{aligned}$$

Then $cl((1\ 2)) = \{(1\ 2), (1\ 3), (2\ 3)\}$.

Theorem 1.3.1: The relation $\sim$ is an equivalence relation on G.

Proof: First, we will show that $\sim$ is reflexive. Note that $x \sim x\ \forall x \in G$ since $x = exe^{-1}\ \forall x \in G$. Next, we need to show that $\sim$ is symmetric. Suppose $x \sim y$. Then we have

$$\begin{aligned}
x \sim y &\implies x = gyg^{-1} \text{ for some } g \in G\\
&\implies g^{-1}x = g^{-1}gyg^{-1}\\
&\implies g^{-1}x = yg^{-1}\\
&\implies g^{-1}xg = yg^{-1}g\\
&\implies g^{-1}xg = y\\
&\implies y = (g^{-1})x(g^{-1})^{-1}\\
&\implies y \sim x,
\end{aligned}$$

so $\sim$ is symmetric. Lastly, we check for transitivity. Suppose $x \sim y$ and $y \sim z$. Then $x = gyg^{-1}$ and $y = hzh^{-1}$ for some $g, h \in G$. Then

$$\begin{aligned} x &= g(hzh^{-1})g^{-1} \\ &= (gh)z(h^{-1}g^{-1}) \\ &= (gh)z(gh)^{-1} \\ &\implies x \sim z, \end{aligned}$$

so $\sim$ is transitive. Then $\sim$ is an equivalence relation on G. $\square$

One last observation that is important to note is that $x \in cl(x)$. Next, we claim that $|cl(x)| = 1$ if and only if $x \in Z(G)$.

Proof: Assume the order of the closure of x is one. Then we have

$$\begin{aligned} |cl(x)| = 1 &\iff \{x\} = cl(x) \\ &\iff \{x\} = \{gxg^{-1} : g \in G\} \\ &\iff x = gxg^{-1} \ \forall g \in G \\ &\iff xg = gxg^{-1}g \ \forall g \in G \\ &\iff xg = gx \ \forall g \in G \\ &\iff x \in Z(G). \ \square \end{aligned}$$

Our big takeaway from this is that conjugacy classes partition the group. Further, if $\{x\}$ is a conjugacy class, then $x \in Z(G)$. In particular, when G is a finite group, it's order can be described as follows:

$$|G| = |Z(G)| + \sum |cl(x)|$$

where the summation is taken over the distinct conjugacy classes of size more than one. The equation above is known as the Class Equation, which we will discuss soon. Note that we can write the class equation in terms of centralizers (subgroups of the form $C_G(x)$).

Example 1.3.8: Consider S_4. Then we have

$$\begin{aligned} |S_4| &= 1 + |cl((1\ 2))| + |cl((1\ 2\ 3))| + |cl((1\ 2\ 3\ 4))| + |cl((1\ 2)(3\ 4))| \\ 24 &= 1 + 6 + 8 + 6 + 3 \end{aligned}$$

Note that for each conjugacy class, it divides the order of the group. That is, $|cl(x)| \big| |G|$.

It is important to note some important facts on the order of groups before we move on. First, if G is a group of even order, then G has an element of order 2. Further, if the order of a group G is the power of a prime (that is, if $|G| = p^n$ with p prime), then $Z(G) \neq 1$.

Definition 1.3.4: Let G be a group with $H \subseteq G$. For $x \in G$, define $xH = \{xh : h \in H\}$ and $Hx = \{hx : h \in H\}$. When $H \leq G$, we call xH a left coset of H in G and Hx a right coset of H in G.

Recall that $G_G(x)\{y \in G : yx = xy\}$. Can we say anything if $aC_G(x) = bC_G(x)$? Consider the following claim: Let $H \leq G$. Then $xH = yH$ if and only if $y^{-1}x \in H$.
Proof: Suppose $xH = yH$. Then $xh_1 = yh_2$ for some $h_1, h_2 \in H$. We have

$$\begin{aligned} xh_1 = yh_2 &\iff x = yh_2h_1^{-1} \\ &\iff y^{-1}x = y^{-1}yh_2h_1^{-1} \\ &\iff y^{-1}x = h_2h_1^{-1} \end{aligned}$$

Note that, since $h_1, h_2 \in H$, we must have that $h_1^{-1} \in H$ and $h_2h_1^{-1} \in H$. Then $y^{-1}x \in H$. $\square$

Next, let A be the set of all left cosets of $C_G(x)$ in G. That is, let $A = \{yC_G(x) : y \in G\}$. Next, define a function $f : A \longrightarrow cl(x)$ by $f(yC_G(x)) = yxy^{-1}$. We claim that f is a bijection.

Proof: Let $axa^{-1} \in cl(x)$. Then $f(aC_G(x)) = axa^{-1}$, so $\exists aC_G(x) \in A$, and thusly f is onto.

Next, suppose you have two equal outputs $f(aC_G(x)) = f(bC_G(x))$. Then we have $axa^{-1} = bxb^{-1}$. But this implies $aC_G(x) = bC_G(x)$, so we must have that f is one-to-one. $\square$.

By an earlier claim, for any $x \in G$, the number of conjugates of x is the same as the number of left cosets of $C_G(x)$ in G. Notationally speaking, for $H \leq G$, we write $[G : H]$ (which reads as the index of H in G), for the number of left cosets of H in G. Then we would say $|cl(x)| = [G : H]$.

This brings our discussion on the order of groups to the Class equation — but first, we must define another theorem.

Theorem 1.3.2: Let G be a group, $x \in G$, and $C_G(x)$ be the centralizer of x in G. Then La Grange's Theorem states that the index of $C_G(x)$ in G is equal to the order of G divided by the order of $C_G(x)$. That is,

$$[G : C_G(x)] = \frac{|G|}{|C_G(x)|}.$$

We will use this to better understand the equation given below.

Definition 1.3.5: Given a finite group G, the Class Equation states that

$$|G| = |Z(G)| + \sum |cl(x)|.$$

That is, the number of elements in G is equal to the number of commutative elements in G plus the sum of the sizes of conjugacy classes containing x where that sum is taken over all conjugacy classes of size more than one with just one x per conjugacy class. Now, we can write this similarly with indeces:

$$|G| = |Z(G)| + \sum [G : C_G(x)],$$

where the sum is taken in the same method as the previous version of the Class Equation.

This brings us to an interest result. We claim that if the order of a group G is equal to the power of a prime p, then there exists some x in the center of G such that x is not the identity element. This brings us to an important theorem.

Theorem 1.3.3: Let p be a prime. If G is a group with $|G = p^k$, $k \in \mathbb{Z}^+$, then $|Z(G)| \geq p$.
Proof: By the class equation, $|G| = |Z(G)| + \sum [G : C_G(x)]$, where the sum is taken over all conjugacy classes of size more than one with one x per conjugacy class. Then, since $|G| = p^k$ and $[G : C_G(x)] = \frac{|G|}{|C_G(x)|}$ by La Grange, we must have

$$p^k = |Z(G) + \sum \frac{p^k}{|C_G(x)|}.$$

Note that, since we are only taking the sum over all conjugacy classes of size mroe than one with one x per conjugacy class, we must have that $|C_G(x)| < p^k$. In other words, if $|C_G(x)| = p^k$, then we must have that $x \in Z(G)$. But we don't want to count any of the x's more than once. Then $\forall x \notin Z(G)$, $C_G(x) \nleq G$. By La Grange, $|C_G(x)||G| = p^k$. Then $|C_G(x)| = p^{l_x}$ for some $0 \leq l_x < k$. This means that each term $\frac{p^k}{|C_G(x)|} = p^{k-l_x}$, which is divisible by p. Then $\sum \frac{p^k}{|C_G(x)|}$ (where the sum is again taken over all conjugacy classes of size more than one with one x per conjugacy class) is the sum of elements that are all divisible by p. Thus, the entire sum is divisible by p. Then we have

$$p^k = |Z(G)| + \sum \frac{p^k}{|C_G(x)|} \text{ and}$$

$$p^k - \sum \frac{p^k}{|C_G(x)|} \text{ divides } p.$$

Thusly, $|Z(G)|\big|p$, so we must have that $|Z(G)| \geq p$ and $Z(G)$ contains a non-identity element. □

1.6 Homomorphisms and Isomorphisms

Definition 1.6.1: Let $(G, \cdot)$ and $(H, \circ)$ be groups. A function $\phi : G \longrightarrow H$ is a homomorphism if

$$\phi(g_1 \cdot g_2) = \phi(g_1) \circ \phi(g_2) \ \forall g_1, g_2 \in G.$$

Note that typically this is written in shorthand as $\phi(g_1 g_2) = \phi(g_1)\phi(g_2)$.

Definition 1.6.2: A bijective homomorphism is called an isomorphism. If there is an isomorphism from G to H, we write $G \cong H$ and say G is isomorphic to H.

Example 1.6.1: Let $(\mathbb{R}, +)$ be the group of real numbers under addition and let $(\mathbb{R}^+, \cdot)$ be the group of positive real numbers under multiplication. Then the function $f : \mathbb{R} \longrightarrow \mathbb{R}^+$ defined by $f(x) = e^x$ is an isomorphism.
To show this, first, we must check that f is a homomorphism. Let $x_1, x_2 \in \mathbb{R}$. Then

$$\begin{aligned} f(x_1 + x_2) &= e^{x_1 + x_2} \\ &= e^{x_1} e^{x_2} \\ &= f(x_1) f(x_2). \end{aligned}$$

Then f is operation preserving and thusly a homomorphism. To check that this is also an isomorphism, we must check that f is a bijection. First, we check that f is one-to-one. Let $f(x_1) = f(x_2)$. Then we have

$$\begin{aligned} f(x_1) = f(x_2) &\implies e^{x_1} = e^{x_2} \\ &\implies \ln e^{x_1} = \ln e^{x_2} \\ &\implies x_1 = x_2, \end{aligned}$$

so f is one-to-one. Next, we check that f is onto. Let $y = f(x)$. Then $y = e^x$ and we have $\ln y = \ln e^x$, so $\ln y = x$. Thusly, we have

$$f(\ln y) = e^{\ln y} = y,$$

so f is onto. Then f is a bijective homomorphism, and thusly, f is an isomorphism.

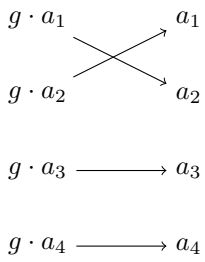
Interestingly, isomorphisms preserve may properties. Assume $\phi : G \longrightarrow H$ is a function with G isomorphic to H. Then we have

1. $|G| = |H|$
2. $ord_G(x) = ord_H(x)$
3. G is abelian $\iff$ H is abelian
4. G is cyclic $\iff$ H is cyclic

Example 1.6.2: Is $(\mathbb{R}, +) \cong (\mathbb{R}\backslash\{0\}, \cdot)$?
No! Consider $(-1)^2 = e \in (\mathbb{R}\backslash\{0\}, \cdot)$. Then that $|-1| = 2$ but $\nexists x \in (\mathbb{R}, +)$ such that $|x| = 2$ and $x \neq e$.

Now, let us look at a particular family of permutations coming from group actions:

Consider the group G and set A with a fixed $g \in G$. Let our group G act on set A via group action $G \times A \longrightarrow A$ be defined by



Then we can associage to g a map $\sigma_g : A \longrightarrow A$ where $\sigma_g(a) = g \cdot a$. It turns out that σ_g is a bijection, and thus it is a permutation on A. **Proof:** Suppose $\sigma_g(a_1) = \sigma_g(a_2)$. Then we must have

$$\begin{aligned} g \cdot a_1 = g \cdot a_2 &\Longrightarrow g^{-1} \cdot (g \cdot a_1) = g^{-1} \cdot (g \cdot a_2) \\ &\Longrightarrow (g^{-1}g) \cdot a_1 = (g^{-1}g) \cdot a_2 \\ &\Longrightarrow e \cdot a_1 = e \cdot a_2 \\ &\Longrightarrow a_1 = a_2. \end{aligned}$$

Then σ_g is one-to-one. Next, let $a \in A$. We want to find $\sigma_g(a) = a$. Consider $g^{-1} \cdot a \in A$. Then

$$\begin{aligned} \sigma_g(g^{-1} \cdot a) &= g \cdot (g^{-1} \cdot a) \\ &= (gg^{-1}) \cdot a \\ &= e \cdot a \\ &= a. \end{aligned}$$

Then σ_g is a bijection and therefore a permutation on A. □

Now, is this map operation preserving? In other words, is this map a homomorphism?

Proof: Remember, group G acts on set A with the action denoted by $g \cdot a$. For $g \in G$, we define a function

$$\sigma_g : A \longrightarrow A$$

by the rule $\sigma_g(a) = g \cdot a$ for $a \in A$. We We proved that σ_g is one-to-one and onto, so it is a permutation of A. Define $\phi : G \longrightarrow S_A$ by the rule $\phi(g) = \sigma_g$. Now, let $g_1, g_2 \in G$.

$\phi(g_1g_2) = \sigma_{g_1g_2}$. Now for any $a \in A$,

$$\begin{aligned} \sigma_{g_1g_2}(a) &= (g_1g_2) \cdot a \\ &= g_1 \cdot (g_2 \cdot a) \\ &= \sigma_{g_1}(g_2 \cdot a) \\ &= \sigma_{g_1}(\sigma_{g_2}(a)) \\ &= \sigma_{g_1} \circ \sigma_{g_2}(a). \end{aligned}$$

This shows that $\sigma_{g_1g_2} = \sigma_{g_1} \circ \sigma_{g_2}$. therefore

$$\begin{aligned} \phi(g_1g_2) &= \sigma_{g_1g_2} \\ &= \sigma_{g_1} \circ \sigma_{g_2} \\ &= \phi(g_1) \circ \phi(g_2). \ \square \end{aligned}$$

Definition 1.6.3: The map ϕ is called the permutation representation of the action. If ϕ is one-to-one (this means distinct elements of G give distinct permutations of A), then the action is said to be faithful.

Definition 1.6.4: The kernel of G acting on a set A is $\{g \in G : g \cdot a = a \ \forall a \in A\}$. This is the same thing as $\ker(\phi) = \{g \in G : \phi(g) = id_A\}$, where id_A is the identity element in S_A.

Example 1.6.3: Look at the cosets of $H =< (1\ 2) >= \{(1\ 2), (1)\}$ in $G = S_3$. Let $A = \{(1), H, (1\ 3)H, (2\ 3)H\}$. Then G acts on A by the rule $\alpha \cdot \beta H = \alpha\beta H$, where $\alpha \in G$, βH is a coset in A, and $\alpha\beta H$ is a coset in A.

(1)H	(1 3)H	(2 3)H
(1 2)	(1 2 3)	(1 3 2)
(1)	(1 3)	(2 3)

Lets try $\alpha = (1\ 2)$. Then we have

$$\begin{aligned}(1\ 2)(1)H &= (1\ 2)H = (1)H \\ (1\ 2)(1\ 3)H &= \{(1\ 2)(1\ 2\ 3), (1\ 2)(1\ 3)\} \\ &= \{(2\ 3), (1\ 3\ 2)\} = (2\ 3)H.\end{aligned}$$

2 Chapter 2: Subgroups

2.1 Subgroups and Examples

Definition 2.1.1: Let G be a group. The subset H of G is a subgroup of G if H is nonempty and H is closed under products and inverses. That is, H is a subgroup of G if

1. $xy \in H \ \forall x, y \in H$ and
2. $x^{-1} \in H \ \forall x \in H$.

If H is a subgroup of G then we write $H \leq G$.

Example 2.1.1: Consider the following examples of subgroups:

1. $\mathbb{Z} \leq \mathbb{Q}$ and $\mathbb{Q} \leq \mathbb{R}$ (both under addition).
2. Any group G has at least two subgroups — $H = G$ and $H = \{e\}$. Note that the latter is called the trivial subgroup (this is denoted by 1 in Dummit and Foote).
3. Let $G = D_{2n}$ be the dihedral group of order $2n$, and let $H = \{1, r, r^2, \ldots, r^{n-1}\}$ be the set of all rotations in G. Since the product of two rotations is again a rotation and the inverse of a rotation is also a rotation, it follows that $H \leq_{2n}$ with $|H| = n$.

Example 2.1.2: Consider the following non-examples of subgroups:

1. $\mathbb{Q}\backslash\{0\}$ under multiplication is not a subgroup of $\mathbb{R}$ under addition, even though both are groups and $\mathbb{Q}\backslash\{0\}$ is a subset of $\mathbb{R}$.
2. $\mathbb{Z}^+ \not\leq \mathbb{Z}$ since $e \notin \mathbb{Z}^+$.
3. $D_6 \not\leq D_8$ since $D_6 \not\subseteq D_8$.

Note that the subgroup relationship is transitive: that is, $H \leq G \leq K \implies H \leq K$.

Proposition 2.1.1: The Subgroup Criterion — A subset H of a subgroup G is a subgroup if and only if

1. $H \neq \emptyset$ and
2. $xy^{-1} \in H \ \forall x, y \in H$.

Example 2.1.3: Let G be a group with $x \in G$, and let $< x >$ be the smallest subgroup of G that contains x. It turns out that $< x >$ can easily be described:

1. Case 1: $|x| = n$ (that is, the order of x is finite). Then $< x >= \{e, x, x^2, \ldots, x^{n-1}\}$, and
2. Case 2: $|x| = \infty$ (that is, the order of x is infinite). Then $< x >= \{\ldots, x^{-2}, x^{-1}, e, x, x^2, \ldots\}$.

Lets prove case 1 — that $H =< x >\leq G$.

Proof: Let $H \leq G$ with $x \in H$. Since H is closed, $x, x^2, x^3, \ldots, x^n \in H$. Then $\{e, x, x^2, \ldots, x^{n-1}\} \subseteq H$. Now we claim that $\{e, x, x^2, \ldots, x^n\} \leq G$. Let $x^i, x^j \in \{e, x, x^2, \ldots, x^{n-1}\}$. Then $i, j \in \{0, 1, 2, \ldots, n-1\}$. Then $x^o x^j = x^{i+j} = x^r$ for some $r = 0, 1, 2, \ldots, n-1^*$. Thus we have closure under products.

Similarly, $(x^i)^{-1} = x^{-i} = x^{r'}$. Then we have shown that $\{e, x, x^2, \ldots, x^{n-1}\}$ forms a subgroup. Further, since any subgroup H containing x must contain $\{e, x, x^2, \ldots, x^{n-1}\}$, we may say that $< x >= \{e, x, x^2, \ldots, x^{n-1}\}$. □

For a quick note on $*$, suppose $|x| = n$. If $m \in \mathbb{Z}$, then $x^m \in \{e, x, x^2, \ldots, x^{n-1}\}$ since $x^m = x^{nq+r} = x^r$.

2.2 Centralizers, Normalizers, Stabilizers, and Kernels

Definition 2.2.1: Let $A \subseteq G$ with G a group. The centralizer of A in G is

$$C_G(A) = \{g \in G : ga = ag \ \forall a \in A\}$$

and the normalizer of A in G is

$$N_G(A) = \{g \in G : gAg^{-1} = A\} \text{ where } gAg^{-1} = \{gag^{-1} : a \in A\}.$$

Note that, if $g \in C_G(A)$, then g commutes with all elements of A, whereas if $g \in N_G(A)$, then $ga_1 = a_2 g$. That is, we might not get back the same element in A after conjugation.

Example 2.2.1: Let $G = D_{2\cdot 4} = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$, and let $A = \{e, r, r^2, r^3\} \leq D_{2\cdot 4}$. Consider that $sr = r^3 s$, so s does not commute with r (similarly with sr, sr^2, sr^3). Then $C_G(A) = A$ (this is strictly a coincidence).

Now, since $C_G(A) \leq N_G(A)$, we know $\{e, r, r^2, r^3\} \subseteq N_G(A)$. Consider sAs^{-1}. We have

$$\begin{aligned} ses^{-1} &= e \\ srs^{-1} &= srs = s^2 r^{-1} = s^2 r^3 \\ sr^2 s^{-1} &= sr^2 s \\ sr^3 s^{-1} &= srs^{-1}. \end{aligned}$$

Then $s \in N_G(A)$.

2.3 Cyclic Groups and Cyclic Subgroups

Theorem 2.3.1: If H is a cyclic group with $K \leq H$, then K is cyclic.
Proof: Let $< x >$ be a generator of H, so $< x >= H$. Let $K \leq H$ and assume $K \neq \{e_H\}$. Let $x^d \in K$ where d is the smallest positive integer with the following claim:

Claim: $< x^d >= K$.

First, since $x^d \in K$ and K is a subgroup, $< x^d > \subseteq K$. Let $y \in K$. Since $y \in H$, we can write $y = x^m$ for some $m \in \mathbb{Z}$. By the division algorithm, we can write $m = dq + r$ with $dq \in \mathbb{Z}$ and $r \in \{0, 1, 2, \ldots, d-1\}$. Then we have

$$\begin{aligned} x^m = x^{dq+r} &\implies x^m x^{-dq} = x^r \\ &\implies x^r \in K. \end{aligned}$$

But, by the way that d was chosen, we must have $r = 0$, so $m = dq$. Thus, we have

$$\begin{aligned} y &= x^m \\ &= x^{dq} \\ &= (x^d)^q \in < x^d > \\ &\implies K \subseteq < x^d > . \square \end{aligned}$$

Example 2.3.1: Consider $\mathbb{Z}_{12} = \{0, 1, 2, \ldots, 11\}$. Then we have

$$\begin{aligned}
< 0 > &= \{0\} \\
< 1 > &= \{0, 1, 2, \ldots, 11\} = \mathbb{Z}_{12} \\
< 2 > &= \{0, 2, 4, 6, 8, 10\} \\
< 3 > &= \{0, 3, 6, 9\} \\
< 4 > &= \{0, 4, 8\} \\
< 5 > &= \{0, 1, 2, \ldots, 11\} = \mathbb{Z}_{12} \\
< 6 > &= \{0, 6\} \\
< 7 > &= \{0, 1, 2, \ldots, 11\} = \mathbb{Z}_{12} \\
< 8 > &= \{0, 4, 8\} \\
< 9 > &= \{0, 3, 6, 9\} \\
< 10 > &= \{0, 2, 4, 6, 8, 10\} \\
< 11 > &= \{0, 1, 2, \ldots, 11\} = \mathbb{Z}_{12}
\end{aligned}$$

Theorem 2.3.2: Let $H =< x >$ with $|x| = n$. For any $m \in \mathbb{Z}$, $< x^m =< x^{\gcd(n,m)} >$

Proposition 2.3.1: Suppose H is a cyclic group with generator $< x >= H$. Then

1. if $|H| = n$, then $|x| = n$ and $H = \{e, x, x^2, \ldots, x^{n-1}\}$
2. if $|H| = \infty$, then $|x| = \infty$ and $H = \{\ldots, x^{-1}, x^{-1}, e, x, x^2, \ldots\}$.

Theorem 2.3.3: If H_1 and H_2 are cyclic groups such that $|H_1| = |H_2|$, then $H_1 \cong H_2$. In particular, if $|H| = n$, then $H \cong \mathbb{Z}_n$ and if $|H| = \infty$, then $H \cong \mathbb{Z}$. We intend to prove the second part of the theorem above.

Proof: Assume $|H| = \infty$, and let $< x >= H$. Define $\phi : H \longrightarrow \mathbb{Z}$ by $\phi(x^i) = i$ with $i \in \mathbb{Z}$. By proposition 2.3.1, since $|H| = \infty$, we have $H = \{\ldots, x^{-2}, x^{-1}, e, x, x^2, \ldots\}$, with each x^i distinct. Next, we must show that ϕ is bijective.

Assume $\phi(x^i) = \phi(x^j)$. Then we have

$$\begin{aligned}
\phi(x^i) = \phi(x^j) &\Longrightarrow i = j \\
&\Longrightarrow x^i = x^j,
\end{aligned}$$

so we must have that ϕ is one-to-one. Next, let $i \in \mathbb{Z}$. Then by definition of ϕ, $\exists x^i \in< x >$ such that $\phi(x^i) = i$, so ϕ is onto. Next, we aim to show that ϕ is operation preserving. We have

$$\begin{aligned}
\phi(x^i x^j) &= \phi(x^{i+j}) \\
&= i + j \\
&= \phi(x^i) + \phi(x^j).
\end{aligned}$$

Then $H \cong \mathbb{Z}$. Now, we consider the second case.

Assume $|H| = n < \infty$. Define $\phi : H \longrightarrow \mathbb{Z}_n$ by $\phi(x^i) = i$ for $i \in \{0, 1, 2, \ldots, n-1\}$. By proposition 2.3.1, $H = \{e, x, x^2, \ldots, x^{n-1}\}$ with each x^i distinct. Next, we must show that ϕ is bijective.

Assume $\phi(x^i) = \phi(x^j)$. Then we have

$$\begin{aligned}
\phi(x^i) = \phi(x^j) &\Longrightarrow i = jk + r \text{ with } k, r \in \mathbb{Z} \text{ and } 0 \leq r \leq n-1 \\
&\Longrightarrow x^i = x^j,
\end{aligned}$$

so we must have that ϕ is one-to-one. Next, let $i \in \mathbb{Z}$. Then by definition of ϕ, $\exists x^i \in < x >$ such that $\phi(x^i) = i$, so ϕ is onto. Next, we aim to show that ϕ is operation preserving. Consider $i, j \in \{0, 1, 2, \ldots, n-1\}$. Then we have

$$\begin{aligned}\phi(x^i x^j) &= \phi(x^{i+j}) \\ &= i + j \text{ if } i + j \in \{0, 1, 2, \ldots, n-1\}.\end{aligned}$$

If $i + j \notin \{0, 1, 2, \ldots, n-1\}$, then we have

$$\begin{aligned}\phi(x^i x^j) &= \phi(x^{i+j} x^{-n}) \\ &= i + j - n \\ &= i + j \\ &= \phi(x^i) + \phi(x^j).\end{aligned}$$

Then $H \cong \mathbb{Z}_n$. That is, every cyclic group is (up to isomorphism) $\mathbb{Z}$ or $\mathbb{Z}_n$. $\square$

Proposition 2.3.2: Let G be a group with $x \in G$ and $m \in \mathbb{Z}\backslash\{0\}$. Then

1. $|x| = \infty \implies |x^m| = \infty$, and
2. $|x| = n \implies |x^m| = {}^{n}/_{\gcd(m,n)}$.

Proof: Assume $|x^m| = l \in \mathbb{Z}^+$. Then $(x^m)^l = x^{ml} = e$. Then consider two cases:

1. $m \geq 1 \implies ml \geq 1$, but $|x| = \infty$, or
2. $m \leq -1 \implies -ml \geq 1$ and $x^{ml} = e \implies x^{-ml} = e$, but $|x| = \infty$.

For part 2, suppose $|x| = n$. First, note that $\gcd(m,n) \big| n$ and $\gcd(m,n) \big| m$. Then we have

$$\begin{aligned}(x^m)^{\frac{n}{\gcd(n,m)}} &= (x^n)^{\frac{m}{\gcd(n,m)}} \\ &= e^{\frac{m}{\gcd(n,m)}} \\ &= e\end{aligned}$$

So $|x^m| \leq \dfrac{n}{\gcd(n,m)}$. Now, suppose $t \in \mathbb{N}$ such that $(x^m)^t = e$. Then we have

$$\begin{aligned}(x^m)^t = e &\implies x^{mt} = e \\ &\implies n \big| mt \text{ since } |x| = n \\ &\implies \frac{n}{\gcd(n,m)} \Bigg| \frac{m}{\gcd(n,m)} \cdot t.\end{aligned}$$

Then note that $\dfrac{n}{\gcd(n,m)}$ is relatively prime to $\dfrac{m}{\gcd(n,m)}$. Then we have that $\dfrac{n}{\gcd(n,m)} \Big| t$, which implies that $\dfrac{n}{\gcd(n,m)} \leq t$. Thusly, we conclude that $|x^m| = \dfrac{n}{\gcd(n,m)}$. $\square$.

Proposition 2.3.3: Let $H =< x >$. Then

1. $|H| = \infty \implies H =< x^m > \iff m = 1$ or $m = -1$, and
2. $|H| = n \implies H =< x^m > \iff \gcd(n,m) = 1$.

Definition 2.3.1: Let $n \in \mathbb{N}$, and let $\phi(n)$ be the number of integers in $\{1, 2, \ldots, n-1\}$ that are relatively prime with n. This is called the Euler Phi Function.

Note that we can use part 2 of the proposition above to prove that the number of generators of $\mathbb{Z}_n$ is $\phi(n)$.

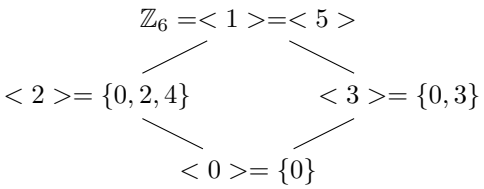
Proposition 2.3.4: Let $H =< x >$. Then

1. $K \leq H \Longrightarrow K$ is cyclic and a generator of K is x^d where d is the smallest positive integer such that $x^d \in K$.
2. $|H| = \infty \Longrightarrow$ for any distinct integers $a, b \geq 0$, $< x^a > \neq < x^b >$ and $< x^m > = < x^{-m} >$.
3. $|H| = n \Longrightarrow$ for each divisor a of n, there is a unique subgroup of H with a elements, and this subgroup is $< x^{n/a} >$. Also, $< x^m > = < x^{\gcd(n,m)} >$.

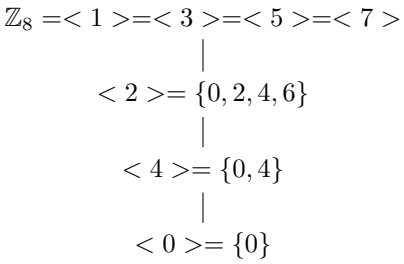
2.5 Subgroup Lattices

This section is a short compilation of examples of subgroup lattices. Lattices are simply graphs associated with groups that can help us to visualize the relationships between groups and their subgroups.

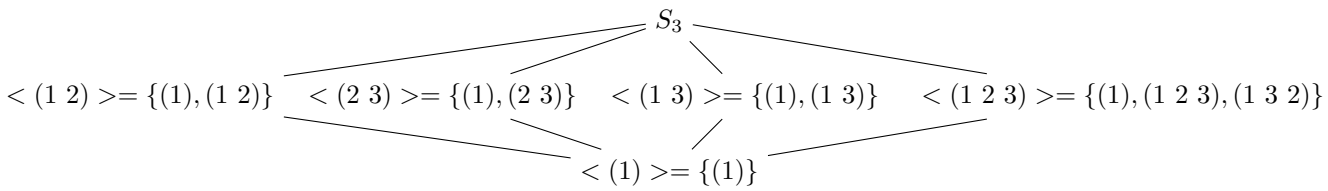
Example 2.5.1: Consier $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$. Then we have



Example 2.5.2: Consider $\mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then we have



Example 2.5.3: Consider $S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Then we have



3 Chapter 3: Quotient Groups and Homomorphisms

3.1 Definitions and Examples

Definition 3.1.1: Let $\phi : G \longrightarrow$ be a group homomorphism. The kernel of this map is $\ker(\phi) = \{g \in G : \phi(g) = e_H\}$.

Proposition 3.1.1: Let $\phi : G \longrightarrow H$ be a group homomorphism. Then

1. $\phi(e_G) = e_H$
2. $\phi(g^{-1} = \phi(g)^{-1})$
3. $\phi(g^n) = \phi(g)^n$
4. $\ker(\phi) \leq G$
5. $Im(\phi) \leq H$ where $Im(\phi) = \{h \in H : \phi(g) = h \text{ for some } g \in G\}$.

Let us sketch a proof for the first statement in the proposition above.

Proof: We have

$$\begin{aligned} e_G \cdot e_G = e_G &\implies \phi(e_G) = \phi(e_G \cdot e_G) \\ &\implies \phi(e_G) = \phi(e_G) \cdot \phi(e_G). \end{aligned}$$

Since $\phi(e_G) \in H$ and $\phi(e_G) = \phi(e_G) \cdot \phi(e_G)$, we must have that $\phi(E_G) = e_H$. $\square$

Definition 3.1.2: For $N \leq G$ and $g \in G$, $gN = \{gn : n \in N\}$ is a left coset of N in G and $Ng = \{ng : n \in N\}$ is a right coset of N in G. An element $x \in gN$ or $x \in Ng$ is called a coset representative.

Now, let G be a group with $N \leq G$. Given any two left cosets of N in G, say g_1N and g_2N, we can combine the cosets by defining $g_1N \cdot g_2N = (g_1g_2)N$, where $\cdot$ is the rule that we are defining and (g_1g_2) is the product of g_1 and g_2 in G. In general, this rule is *not* an operation.

Definition 3.1.3: $N \leq G$ is a normal subgroup if $gN = Ng \ \forall g \in G$. Equivalently, $gNg^{-1} = N \ \forall g \in G$. Notationally, we say N is a normal subgroup of G by saying $N \lhd G$. In terms of $N_G(N)$ (the normalizer of N in G), N is normal in G if and only if $N_G(N) = G$.

Theorem 3.1.1: Let G be a group with $N \leq G$. Then the following are equivalent:

1. $N \trianglelefteq G \iff gNg^{-1} = N \ \forall g \in G$
2. $N_g(N) = G$
3. $gN = Ng \ \forall gG$
4. $g_1N \star g_2N := (g_1g_2)N$ utilizes a well-defined operation on the set of left cosets of N in G
5. $gNg^{-1} \subseteq N \ \forall g \in G$

We will utilize a chain of proofs for the theorem above. First, we prove that the first item implies the second.
Proof ($1 \implies 2$): Let G be a group with $N \leq G$. Assume $N \trianglelefteq G$. Then, by definition, $gNg^{-1} = N \ \forall g \in G$. Therefore, $g \in G \implies g \in N_G(N)$, so $N_G(N) = G$. $\square$

Proof ($2 \implies 3$): Assume $N_G(N) = G$. First, we claim that $gN \subseteq Ng$. Let $gn \in gN$. Since $g \in N_G(N)$, $gng^{-1} \in N$. Then $gng^{-1} = n_2 \in N$. We have $gng^{-1} = n_2 \implies gn = n2g \in Ng$, then $gN \subseteq Ng$.

Next, we claim that $Ng \subseteq gN$. In doing so, we claim $g^{-1} \in N_G(N)$. We have

$$\begin{aligned} g^{-1} \in N_G(N) &\implies g^{-1} \in G : g^{-1}N(g^{-1})^{-1} = N \\ &\implies g^{-1}n_1 = n_2 g^{-1} \\ &\implies gg^{-1}n_1 = gn_2 g^{-1} \\ &\implies n_1 = gn_2 g^{-1} \\ &\implies n_1 g = gn_2 g^{-1} g \\ &\implies n_1 g = gn_2 \end{aligned}$$

Now, note that $n_1 g \in Ng$ and $gn_2 \in gN$, so $n_1 g = gn_2 \implies n_1 g \in gN \implies Ng \subseteq gN$. $\square$

Proof ($3 \implies 4$): Assume $gN = Ng \ \forall g \in G$. Suppose $g_1 N = g_2 N$ and $h_1 N = h_2 N$. We need to show $(g_1 h_1)N = N(g_2 h_2)$. Note that

$$\begin{aligned} g_1 N = g_2 N &\implies \exists n_1 \in N : g_2^{-1} g_1 = n_1 \text{ and} \\ h_1 N = h_2 N &\implies \exists n_2 \in N : h_2^{-1} h_1 = n_2. \end{aligned}$$

Solving for g_1 and h_1 gives us $g_1 = g_2 n_1$ and $h_1 = h_2 n_2$. Then we have

$$\begin{aligned} (g_1 h_1)N &= (g_2 n_1 h_2 n_2)N \\ &= g_2 n_1 h_2 n_2 N \\ &= g_2 n_1 h_2 N \\ &= g_2 h_2 n_3 N \\ &= g_2 h_2 N \\ &= (g_2 h_2)N \end{aligned}$$

Proof ($4 \implies 5$): Assume $g_1 N \star g_2 N = (g_1 g_2)N$, with $\star$ on left cosets being well-defined. We want to show $gNg^{-1} \subseteq N \ \forall g \in G$. We have

$$\begin{aligned} gng^{-1}N &= (gnN)(g^{-1}N) \\ &= (gN)(g^{-1}N) \\ &= (gg^{-1})N \\ &= eN \\ &= N \end{aligned}$$

so we must have that $gng^{-1} \in N \ \forall g \in G \ \forall n \in N$, and thusly $gNg^{-1} \subseteq N \ \forall g \in G$. $\square$

Proof ($5 \implies 1$): Assume $gNg^{-1} \subseteq N \ \forall g \in G$. Then we need to show $N \subseteq gNg^{-1} \ \forall g \in G$. Let $n \in N$. We need to show $n \in gNg^{-1}$. We have

$$\begin{aligned} n &= (gg^{-1})n(gg^{-1}) \\ &= g(g^{-1}ng)g^{-1} \\ &= g(g^{-1}n(g^{-1})^{-1})g^{-1} \\ &= gn_2 g^{-1} \text{ for some } n_2 \in N \end{aligned}$$

so $n \in gNg^{-1} \ \forall g \in G$. Thusly, $N \subseteq gNg^{-1}$ and therefore $gNg^{-1} = N \ \forall g \in G$. $\square$

As a consequence, we can now define quotient groups.

Definition 3.1.4: We say that if $N \trianglelefteq G$, then we have a well-defined operation G/N (the set of all left cosets of N in G). In fact, G/N forms a group under this operation. This group is called a quotient group or factor group.

Proposition 3.1.2: Let $N \leq G$. Then $N \trianglelefteq G$ if and only if $N = \ker(\phi)$ with ϕ a homomorphism.

Proof: Assume $N \trianglelefteq G$. Then define the map $\pi : G \longrightarrow {}^{G}/_{N}$ by $\pi(g) = gN$. Then, for any $g_1, g_2 \in G$, we have

$$\begin{aligned}\pi(g_1 g_2) &= g_1 g_2 N \\ &= (g_1 N)(g_2 N) \\ &= \pi(g_1)\pi(g_2).\end{aligned}$$

Now

$$\begin{aligned}g \in \ker(\pi) &\iff \pi(g) = eN \\ &\iff gN = N \\ &\iff g \in N,\end{aligned}$$

thus $\ker(\pi) = N$.

Now, for the other direction, assume N is the kernel of some group homomorphism $g : G \longrightarrow K$. We know $N \leq G$. Let $g \in G,\ n \in N$. We need to find $gng^{-1} \in N$. We have

$$\begin{aligned}f(gng^{-1}) &= g(g)f(n)f(g)^{-1} \\ &= f(g)e_K f(g)^{-1} \\ &= e_K.\end{aligned}$$

This shows $gng^{-1} \in \ker(f) = N\ \forall g \in G\ \forall n \in N$. Then $gNg^{-1} \subseteq N\ \forall g \in G$. Thusly, $N \trianglelefteq G$. $\square$

Next, we consider a vital theorem, the First Isomorphism Theorem.

Theorem 3.1.2: The First Isomorphism Theorem
If $\phi : G \longrightarrow H$ is a group homomorphism, then $\ker(\phi) \trianglelefteq G$ and ${}^{G}/_{\ker(\phi)} \cong \phi(G)$, where $\phi(G) = Im(G)$.

Proof: We have shown (above) that $\ker(\phi) \trianglelefteq G$. Then we need to show ${}^{G}/_{\ker(\phi)} \cong \phi(G)$. Define

$$\psi : {}^{G}/_{\ker(\phi)} \longrightarrow Im(G)$$

by the rule $\psi(g\ker(\phi)) = \phi(g)$. Let $\ker(\phi) = K$. We then need to show that ψ is an isomorphism. First, we need to see if ψ is well-defined. Suppose $g_1 K = g_2 K$. Then we have $g_2^{-1} g_1 \in K$. Since $K = \ker(\phi)$, we have $\phi(g_2^{-1} g_1) = e_H$. This can be rewritten as $\phi(g_1) = \phi(g_2)$. Then ψ is well-defined (and one-to-one, which can be easily shown by reversing the well-defined argument above). Next, let $h \in Im(\phi)$, so $h = \phi(g)$ for some $g \in G$. Then $\psi(gK) = \phi(g) = h$. To check for operation preservation, consider

$$\begin{aligned}\psi(g_1 K g_2 K) &= \psi(g_1 g_2 K) \\ &= \phi(g_1 g_2) \\ &= \phi(g_2)\phi(g_2) \\ &= \psi(g_1 K)\psi(g_2 K).\ \square\end{aligned}$$

Theorem 3.1.3: Let G be a finite group with $H \leq G$. If the number of left cosets of H in G is p with $p \big| |G|$ and p prime, then $H \trianglelefteq G$.
Proof: Let G act on the set of elft cosets of H in G by the rule $g \cdot aH = gaH$. This gives a homomorphism from G to S_p. By the first isomorphism theorem, we can say

$${}^{G}/_{\ker(\phi)} \cong Im(\phi).$$

Now $Im(\phi) \leq S_p$. By La Grange, $|Im(\phi)| \big| p!$ (as $|S_p| = p!$). Since $\left|{}^{G}/_{\ker(\phi)}\right| = |Im(\phi)|$, we have that

$$\begin{aligned}{}^{|G|}/_{|\ker(\phi)|} p! &\implies {}^{|G|}/_{|\ker(\phi)|} \Big| p \\ &\implies {}^{|G|}/_{|\ker(\phi)|} = 1 \text{ or } {}^{|G|}/_{|\ker(\phi)|} = p.\end{aligned}$$

If $|G|/|\ker(\phi)| = 1$, then $G = \ker(\phi)$, which would give that $\phi(g) = id_{S_p}\ \forall g \in G$. But then for any $g \notin H$, $gH \neq H$ so that $\sigma_g \neq id_{S_p}$. Then, because of the contradiction, we must have that $|G|/|\ker(\phi)| = p$.

Notice that $g \in \ker(\phi)$, so $\phi(g) = \sigma_g$ is the identity permutation in S_p. In particular, $gH = H$, which holds if and only if $g \in H$. Then we must have that $\ker(\phi) \cong H$.

Then $|G|/|H| \leq |G|/|\ker(\phi)| = p$. Since there are p left cosets of H in G, we also have that $p = |G|/|H|$. Then $|H| = |\ker(\phi)|$ and since $\ker(\phi) \subseteq H$, we have that $\ker(\phi) = H$. Since $\ker(\phi) \trianglelefteq G$, we have that $H \trianglelefteq G$. $\square$

One consequence of this result is if G is finite, H is a subgroup of G, and there are only two left cosets of H in G, then $H \trianglelefteq G$.

3.2 More on Cosets and La Grange's Theorem

Theorem 3.2.1: La Grange's Theorem
If G is finite with $H \leq G$, then $|H|\big||G|$, and there are $|G|/|H|$ left cosets of H in G.

Definition 3.2.1: For $H \leq G$, the index of H in G, denoted $[G : H]$, is the number of left cosets of H in G. If G is finite, then $[G : H] = |G|/|H|$.

Corollary 3.2.1: If G is finite and $x \in G$, then $|x|\big||G|$.
Proof: Observe $|< x >| = |x|$. Then, by La Grange, $|< x >|\big||G|$. $\square$

Corollary 3.2.2: If $|G| = p$ with p prime, then $G \cong \mathbb{Z}_p$.
Proof: Let $x \in G$ with $x \neq e$. Then $|< x >| > 1$ and $|< x >| = p$. Then $< x >= G$. $\square$

Definition 3.2.2: Let G be a group with $H, K \leq G$. Define $HK = \{hk : h \in H, k \in K\}$.

Proposition 3.2.1: If $H, K \leq G$ with H and K both finite, then $|HK = \dfrac{|H| \cdot |K|}{|H \cap K|}$.
Proof: We can write $HK = \bigcup_{h \in H} hK$. Note that two cosets of the form h_1K and h_2K are equal if and only if $h_2^{-1}h_1 \in K$. Now, since $H \leq G$, $h_2^{-1}h_1 \in H$. Then

$$\begin{aligned} h_1K = h_2K &\iff h_2^{-1}h_1 \in H \cap K \\ &\iff h_1(H \cap K) = h_2(H \cap K). \end{aligned}$$

This means that, in the union $HK = \bigcup_{h \in H} hK$, there are $|H \cap K|$ cosets hK that are all the same coset. Then $\left|\bigcup_{h \in H} hK\right| = \dfrac{|H|}{|H \cap K|}|K|$. $\square$

3.4 Composition Series and the Hölder Program

Proposition 3.4.1 If G is a finite abelian group and p is a prime such that $p\big||G|$, then $x \in G$ such that $ord(x) = p$.
Proof: We will prove by induction. First, assume p is a prime such that $p\big||G|$. Then let the base case assume that $|G| = p$. In this case, if $x \in G$ with $x \neq e$. Then $|x| \neq 1$ and $|x|\big|p$ by La Grange. Since p is prime, $|x| = p$.

Next, for the inductive step, assume the base case result holds now for all abelian groups with less than $|G|$ elements. That is, assume that if H is an abelian group and $p\big||H|$ with $|H| < |G|$, then $h \in H$ with $|h| = p$. Let $x \in G$ with $x \neq e$. If $p\big||x|$, say $|x| = pk$ with $k \in \mathbb{Z}$, then

$$|x^k| = \frac{pk}{k} = p \Longrightarrow |x^k| = p$$

and we found an element of order p. Next, assume $p \nmid |x|$. Consider $N =< x >$ is a subgroup. Since G is abelian, we know $N \trianglelefteq G$ and $p \nmid |N|$. Lets look at the abelian group G/N. Notice $|G/N| = |G|/|N| < |G|$ since $x \neq e$. Also, $p\big||G|/|N|$ since $p||G|$ but $p \nmid |N|$. By the inductive hypothesis, G/N has an element of order p,

say yN in G/N. However, the map $\pi : G \longrightarrow G/N$ defined by $\pi(g) = gN$ is a group homomorphism. Thus $|yN|_{G/N} \Big| |y|_G \Longrightarrow p \Big| |y|_G$, and now we can apply the first part of the argument if $|y|_G = pl$, then $|y^l|_G = p$. □

Definition 3.4.1: Let G be a group. Then G is called simple if its only normal subgroups are $\{e\}$ and G.

3.5 Transpositions and the Alternating Groups

Definition 3.5.1: In the permutation group, a cycle of length 2 is called a transposition.

Consider now a cycle $(a_1\ a_2\ \ldots\ a_m)$ of length m. Any cycle can be written as a product (composition) of $m-1$ transpositions.

Cycle Length	**Product of Transpositions**
2 so $(a\ b)$	$(a\ b)$
3 so $(a\ b\ c)$	$(a\ c)(a\ b)$
4 so $(a\ b\ c\ d)$	$(a\ d)(a\ c)(a\ b)$
$\vdots$	$\vdots$
m so $(a_1\ a_2\ \ldots\ a_m)$	$(a_1\ a_m)(a_1\ a_{m-1})\ldots(a_1\ a_3)(a_1\ a_2)$

Since every permutation in S_n can be written as a product of disjoint cycles, and each cycle can be written as a product of transpositions, every $\sigma \in S_n$ is a product of transpositions. Note that they need not be disjoint.

Example 3.5.1: Consider

$$\begin{aligned}\sigma &= (1\ 3\ 7\ 9)(2\ 6\ 4)(5\ 8)\\ &= (1\ 9)(1\ 7)(1\ 3)\end{aligned}$$

Definition 3.5.2: We call a permutation even if it can be written as a product of an even number of transpositions. Else, it is odd.

Example 3.5.2: Note that determinants of matrices relate to even and odd permutations. Consider the matrix and its determinant below:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1b_2c_3 + b_1c_2a_3 + c_1a_2b_3$$

We will have more of this in Galois Theory.

Theorem 3.5.1: In S_n, any time the identity permutation is written as a product of transpositions, the number of transpositions must be even.

Theorem 3.5.2: If $\sigma \in S_n$ and $\sigma = \alpha_1\alpha_2\cdots\alpha_r = \beta_1\beta_2\cdots\beta_t$, where each α_i, β_j is a transposition, then r and t are either both even or both odd.

Consider the sketch of this proof below:

$$\begin{aligned} id &= \sigma\sigma^{-1}\\ &= \alpha_1\alpha_2\cdots\alpha_r(\beta_1\beta_2\cdots\beta_t)^{-1}\\ &= \alpha_1\alpha_2\cdots\alpha_r\beta_t^{-1}\beta_{t-1}^{-1}\cdots\beta_1^{-1}\\ &= \alpha_1\alpha_2\cdots\alpha_r\beta_t\beta_{t-1}\cdots\beta_1. \end{aligned}$$

Note that the final line above expresses the identity permutation id as a total of $r+t$ transpositions, so $r+t$ must be even. □

Definition 3.5.3: The Alternating Group of degree n, denoted A_n, is the subgroup of S_n consisting of all even permutations in S_n. Note some interesting properties of A_n below:

1. $A_n \trianglelefteq S_n$
2. $|A_n| = {}^{n!}/_{2}$
3. $[S_n : A_n] = 2$

Example 3.5.3: Consider the following symmetric groups and their corresponding alternating groups. Note that the symmetric groups shown do not list the members of their corresponding alternating groups.

A_2	$S_2 \backslash A_2$
id	$(1\ 2)$

A_3	$S_3 \backslash A_3$
id	$(1\ 2)$
$(1\ 2\ 3)$	$(1\ 3)$
$(1\ 3\ 2)$	$(2\ 3)$

Theorem 3.5.3: Cauchy's Theorem
If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p. Equivalently, G has a subgroup with p elements. Note that we proved this already with abelian groups (see proposition 3.4.1).
Proof: We will use induction on the number of elements in G. Let $|G| = n$. Since $p \big| |G|$, we know $p|n$, so $p \leq n$. This means our base case is $n = p$.

So $|G| = p$ and any non-identity element in G has order p by La Grange's Theorem. Then our induction hypothesis supposes that $n > p$ and $p|n$. Assume Cauchy's Theorem is true for all groups of order m where $m < n$ and $p|m$.

If there is a $y \in G$ such that $p \big| |y|$, say $|y| = kp$ for some $k \in \mathbb{N}$. Then we have $|y^k| = p$ and we are done. Assume that no element has order p. We will obtain a ctonradiction.

Case 1: G is not abelian. By the class equation, we have

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)]$$

where $g_1, g_2, \ldots, g_r$ are representatives from the conjugacy classes of size bigger than one. Since $p \big| |G|$, if $p \big| |Z(G)|$, then since $Z(G) \lneq G$, we can apply our induction hypothesis to $Z(G)$ to get an element $z \in Z(G)$ of order p, which has order p in G.

Assume $p \nmid |Z(G)|$. Then there must exist g_1 such that $p \nmid [G : C_G(g_i)]$. Then $p \nmid {}^{|G|}/_{|C_G(g_i)|}$. But $p \big| |G|$, so $p \big| |C_G(g_i)|$. Now $C_G(g_i) \lneq G$ and $p \big| |C_G(g_i)|$. Then, by our induction hypothesis, $C_G(g_i)$ has an element of order p. $\square$

4 Group Actions

4.4 Automorphisms

Definition 4.4.1: Let G be a group and $\phi : G \longrightarrow G$ be the automorphism $\phi_g(x) = gxg^{-1}$. Let $Inn(G) = \{\phi_g : g \in G\}$ be the inner automorphisms of G. This forms a subgroup $Aut(G)$, the set of all automorphisms of G under function composition.

This idea of using conjugation to generate an automorphism can be generalized into subgroups:

Let $H \leq G$ and $g \in G$. For ϕ_g to be a function from $H \longrightarrow H$, we need

$$\begin{aligned} \phi_g(h) \in H \ \forall h \in H &\iff ghg^{-1} \in H \ \forall h \in H \\ &\iff g \in N_G(H) \end{aligned}$$

Define $\psi : N_G(H) \longrightarrow Aut(H)$ by the rule $\psi(g) = \phi_g$ ($\phi_g(h) = ghg^{-1}$). Note that ψ is a homomorphism:

$$\begin{aligned} \psi(g_1 g_2) &= \phi_{g_1 g_2}(h) \\ &= \phi_{g_1} \circ \phi_{g_2}(h). \end{aligned}$$

By the first isomorphism theorem, we have

$$N_G(H)/\ker(\psi) \cong Im(\psi) \leq Aut(H).$$

Now,

$$\begin{aligned} g \in N_G(H) \text{ is in } \ker(\psi) &\iff \psi(g) = \mathrm{id}_H \text{ with } \mathrm{id}_H(h) = h \ \forall h \in H \\ &\iff \phi_g = \mathrm{id}_H \\ &\iff \phi_g(h) = h \ \forall h \in H \\ &\iff ghg^{-1} = h \ \forall h \in H \\ &\iff gh = hg \ \forall h \in H \\ &\iff g \in C_G(H). \end{aligned}$$

Thus, $\ker(\psi) = C_G(H)$ and we have

$$N_G(H)/C_G(H) \cong Im(\psi) = \{\phi_g : g \in N_G(H)\}.$$

Now, lets return to the case when $H = G$.

$$\begin{aligned} N_G(G) = G \text{ and } C_G(G) = Z(G) \text{ and } Im(\psi) &= \{\psi(g) : g \in G\} \\ &= \{\phi_g : g \in G\} \\ &= Inn(G), \end{aligned}$$

which gives us that $G/Z(G) \cong Inn(G)$. This is a good result to add to our toolbelt when investigating groups.

4.5 Sylow's Theorems

Definition 4.5.1: Let G be a group and p be a prime.

1. A group of order p^α, $\alpha \in \mathbb{Z}^+$, is called a p-group. Subgroups of G that are p-groups are called p-subgroups.
2. If $|G| = p^\alpha m$ with $p \nmid m$, then $H \leq G$ with $|H| = p^\alpha$ is called a p-Sylow subgroup of G.

Example 4.5.1: Consider $\mathbb{Z}_{36}$. We want to find the 3-subgroups of $\mathbb{Z}_{36}$. That is, we are looking for subgroups of order $3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81, \ldots$, etc.

$$< 12 > = \{0, 12, 24\}$$
$$< 4 > = \{0, 4, 8, 12, 16, 20, 24, 28, 32\}.$$

Note that $< 12 > \leq \mathbb{Z}_{36}$ has order 3^1 and $< 4 > \leq \mathbb{Z}_{36}$ has order 3^2. Then $< 12 >$ and $< 4 >$ are both p-subgroups of $\mathbb{Z}_{36}$. Further, since $|\mathbb{Z}_{36}| = 36 = 3^2 \cdot 4$ and $3 \nmid 4$, and since $| < 4 > | = 3^2$ with $< 4 > \leq \mathbb{Z}_{36}$, we have that $< 4 >$ is a 3-Sylow subgroup of $\mathbb{Z}_{36}$.

Example 4.5.2: Consider the symmetric group S_3. Note that the 2-Sylow subgroups in S_3 are $\{id, (1\ 2)\}, \{id, (1\ 3)\}$, and $\{id, (2\ 3)\}$. Further, the only 3-Sylow subgroup in S_3 is $\{id, (1\ 2\ 3), (1\ 3\ 2)\}$.

Note that we write $Syl_p(G)$ to represent the Sylow p-subgroups in G and we write n_p or $n_p(G)$ to represent the number of p-Sylows in G.

Example 4.5.3: In S_3, we have

$$Syl_2(S_3) = \{< (1\ 2) >, < (1\ 3) >, < (2\ 3) >\} \text{ and } n_2 = n_2(S_3) = 3$$
$$Syl_3(S_3) = \{< (1\ 2\ 3) >\} \text{ and } n_3 = n_3(S_3) = 1.$$

Theorem 4.5.1: Sylow's Theorem
Let G be a group and suppose $|G| = p^\alpha m$, where $\alpha \in \mathbb{Z}^+$, p is prime, and $p \nmid m$. Then

1. $Syl_p(G) \neq \emptyset$. That is, G has at least one p-Sylow subgroup.
2. If $Q \leq G$, Q is a p-group, and $P \in Syl_p(G)$, then there exists some $g \in G$ such that $Q \leq gPg^{-1}$. In particular, if $P, Q \in Syl_p(G)$, then there exists some $g \in G$ such that $Q = gPg^{-1}$.
   To put this more simply, any two p-Sylows are conjugates of each other.
3. If n_p is the number of p-Sylows, then $n_p \equiv 1 (\mod p)$ and $n_p | m$. Note that $n_p = [G : N_G(P)]$ where $P \in Syl_p(G)$.

Corollary 4.5.1: Let G be a finite group with $P \in Syl_p(G)$. Then $Syl_p(G) = \{P\}$ if and only if $P \trianglelefteq G$. That is, the number of p-Sylows in G is 1 if and only if P is a normal subgroup of G.

Proof: Assume that $n_p = 1$. That is, assume $Syl_p(G) = \{P\}$. Then, for any $x \in G$, $xPx^{-1} \in Syl_p(G)$ since $xPx^{-1} \leq G$ and $|xPx^{-1}| = |P| = p^\alpha$. Thus, $xPx^{-1} \in Syl_p(G) = \{P\}$ $\forall x \in G$. Then $xPx^{-1} = P$ $\forall x \in G$ and $P \trianglelefteq G$.

For the other direction, assume $P \trianglelefteq G$. Take any $Q \in Syl_p(G)$. Since any two p-Sylows are conjugates (part 2 of Sylow's theorems), $\exists x \in G$ such that $xPx^{-1} = Q$. Then, since $P \trianglelefteq G$, we have $xPx^{-1} = Q \implies P = Q$. $\square$

Definition 4.5.2: The Normal Factoring Lemma
Let G be a finite group with $H, K \trianglelefteq G$, $HK = G$, and $g = hk$ with $h \in H$ and $k \in K$ a unique product in G $\forall g \in G$. Then

1. $H \cap K = \{e\}$, and
2. $G \cong H \times K$.

Next, we consider some applications of Sylow's theorems. The first application relates to groups of order pq. Let G be a group with $|G| = pq$, $p < q$, and $p \nmid q - 1$. Then $G \cong \mathbb{Z}_{pq}$.
Proof: First, let's consider what we know:

1. $n_p \equiv 1(\mod p)$
2. $n_p | q \implies n_p \in \{1, q\}$
3. $n_q \equiv 1(\mod p)$, and
4. $n_q | p \implies n_q \in \{1, p\}$.

Now, consider $n_p = q$. Then we have

$$\begin{aligned} n_p = q &\implies q \equiv 1(\mod p) \\ &\implies p | q - 1. \end{aligned}$$

But this is a contradiction, so we must have that $n_p = 1$. Next, if $n_q = p$, we have

$$\begin{aligned} n_q = p &\implies p \equiv 1(\mod q) \\ &\implies Q | P - 1 \\ &\implies qk = p - 1. \end{aligned}$$

But $p < q$, so this is a contradiction. Then $n_q = 1$. Now, let $P, Q \in Syl_p(G)$. Then $P, Q \trianglelefteq G$ since $n_p = n_q = 1$. Also, $P \cap Q = \{e\}$ since $\gcd(|P|, |Q|) = 1$. Finally,

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = \frac{pq}{1} = pq = |G|.$$

Then, by the normal factoring lemma, $G \cong P \times Q \cong \mathbb{Z}_p \times \mathbb{Z}_q \cong \mathbb{Z}_{pq}$. $\square$

Example 4.5.4: Prove that a group of order 255 is abelian.
First, note that the prime factorization of 255 is $255 = 3 \cdot 5 \cdot 17$.

Then, by Sylow's theorems, $n_3 \equiv 1(\mod 3)$ and $n_3 | 5 \cdot 17$. But $5 \cdot 17 = 85$ and the only factors of 85 are $1, 5, 17$, and 85. So we check our next prime, 5. We have $n_5 \equiv 1(\mod 5)$ and $n_5 | 3 \cdot 17 = 51$. Similarly to the first prime factor we checked, the only factors of 51 are $1, 3, 17$, and 51. Then we must check our final prime factor, 17.

We know, again by Sylow's theorems, $n_{17} \equiv 1(\mod 17)$ and $n_{17} | 3 \cdot 5 = 15$. Note that the divisors of 15 are $1, 3, 5$, and 15. Note that

$$1(\mod 17) \cong \overline{1} = \{\dots, -16, 1, 18, 35, \dots\}.$$

Remember that n_{17} represents the number of p-Sylows, so it must be a positive number. Further, it must divide the product of the other two factors of 255 (15), so it has to be less than or equal to 15. Then we must have that $n_{17} = 1$.

Then let $P \in Syl_{17}(G)$. Then $P \trianglelefteq G$. Since $|P| = 17$, $P \cong \mathbb{Z}_{17}$. By La Grange's theorem, $\left| {}^{N_G(P)}/_{C_G(P)} \right| \Big| |AutP$.
Further, since $P \trianglelefteq G$, we know $N_G(P) = G$. Also, $Aut(P) \cong Aut(\mathbb{Z}_{17}^\star)$. Also, note that $|\mathbb{Z}_{17}^\star| = \phi(17) = 16$.
Now we have

$$\begin{aligned} \left| {}^{G}/_{C_G(P)} \right| 16 &\implies {}^{|G|}/_{|C_G(P)|} \Big| 16 \\ &\implies {}^{3 \cdot 5 \cdot 17}/_{|C_G(P)|} \Big| 16 \\ &\implies |C_G(P)| = 3 \cdot 5 \cdot 17 = |G| \\ &\implies C_G(P) = G \\ &\implies C_G(P) = G \\ &\implies P \leq Z(G). \end{aligned}$$

By La Grange, $17 \big| |Z(G)|$, so we must have that $G/Z(G)$ has either $1, 3, 5$, or $3 \cdot 5$ elements, and $3 \nmid 5 - 1$. In all cases, G is abelian.

Example 4.5.5: Show that a group of order 84 cannot be simple. *Hint: to prove a group is not simple, find a Sylow p-subgroup of the group. Further, start with the bigger primes and work your way down.*

Let $|G| = 84 = 2^2 \cdot 3 \cdot 7$. First, we know $n_2 \equiv 1 (\mod 2)$ and $n_2 | 3 \cdot 7$. Then we must have that $n_2 \in \{1, 3, 7, 21\}$. Similarly, $n_3 \equiv 1 (\mod 3)$ and $n_3 | 2^2 \cdot 7$, so $n_3 \in \{1, 4, 7\}$. Lastly, we know $n_7 \equiv 1 (\mod 7)$ and $n_7 | 2^2 \cdot 3$, so $n_7 \in \{1\}$. Now, since there is only one 7-Sylow subgroup of G (say $P \in Syl_y(G)$), we must have that $P \trianglelefteq G$. $\square$

Example 4.5.6: Show that a group of order 351 cannot be simple.

We will solve this similarly to the previous example. Let $|G| = 351 = 3^3 \cdot 13$. Note that $n_{13} \equiv 1 (\mod 13)$ and $n_{13} | 3^3$. Then we must have that $n_{13} \in \{1, 27\}$. Now, if $n_{13} = 1$, then there is only one 13-Sylow subgroup which will be normal (in this case we would be done). Then assume $n_{13} = 27$. That is, assume there are 27 subgroups of G with cardinality 13. Remember, if $Q =< x >$ with $| < x > | = 13$ (a prime order), then all non-identity elements of Q generate all of Q.

Note that if Q_1 and Q_2 are two distinct subgroups of cardinality 13, then $Q_1 \cap Q_2 = \{e\}$.

Now, since we are assuming that there are 27 subgroups of cardinality 13, every element in each subgroup *except* the identity element will generate the whole group. Then $27 \cdot 12 = 324$. But the order of G is 351, which leaves 27 ($27 = 351 - 324$) elements that are not of order 13.

Since exactly 27 elements remain, and since a 3-Sylow subgroup will have exactly $3^3 = 27$ elements, the remaining elements form a unique 3-Sylow subgroup which is normal in G. $\square$

Example 4.5.7: What are the possible isomorphism classes of a group of order 45?

Let $|G| = 45 = 3^2 \cdot 5$. First, note that $n_3 \equiv 1 (\mod 3)$ and $n_3 | 5$, so $n_3 = 1$. Similarly, $n_5 \equiv 1 (\mod 5)$ and $n_5 | 3^2$, so $n_5 = 1$. Let $P \in Syl_3(G)$ and $Q \in Syl_5(G)$. Then $P, Q \trianglelefteq G$, $P \cap Q = \{e\}$, and $|PQ = \dfrac{|P||Q|}{|P \cap Q|} = \dfrac{3^2 \cdot 5}{1} = 45$, so we must have that $PQ = G$. Then, by the Normal Factoring Lemma, $G \cong P \times Q$.

Now, since $|P| = 3^2$, $P \cong \mathbb{Z}_9$ or $P \cong \mathbb{Z}_3 \times \mathbb{Z}_3$. Similarly, $|Q| = 5$, so $Q \cong \mathbb{Z}_5$. Then $G \cong \mathbb{Z}_9 \times \mathbb{Z}_5$ or $G \cong \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$. $\square$

Example 4.5.8: If $|G| = p^2 q^2$ with p and q distinct primes and $p < q$, then G has a normal q-Sylow unless $|G| = 36$.

First, note that $n_p \equiv 1 (\mod p)$ and $n_p | q^2$, so we must have that $n_p \in \{1, q, q^2\}$. Similarly, $n_q \equiv 1 (\mod q)$ with $n_q | p^2$, so $n_q \in \{1, p, p^2\}$. Now, if $n_q = 1$, then we are done. Then assume $n_q = p$. Then we have

$$\begin{aligned} n_q = p &\implies p \equiv 1 (\mod q) \\ &\iff q | p - 1 \\ &\iff p > q. \end{aligned}$$

But $p < q$, so we have a contradiction.

Then assume $n_q = p^2$. Then we have

$$\begin{aligned} n_q = p^2 &\implies p^2 \equiv 1 (\mod q) \\ &\implies p^2 - 1 \equiv 0 (\mod q) \\ &\implies (p+1)(p-1) \equiv 0 (\mod q) \\ &\implies q | p + 1 \text{ or } q | p - 1 \\ &\implies p > q. \end{aligned}$$

But, as above, this is a contradiction. Since $p < q$ are both distrinct primes and $q | p + 1$, p and q must be consecutive primes. Then $p = 2$ and $q = 3$, so $p^2 q^2 = 4 \cdot 9 = 36$. $\square$

7 Introduction to Rings

7.1 Basic Definitions and Examples

Definition 7.1.1: A ring is a set with two binary operations $+$ and $\times$, called addition and multiplication, such that

1. $(R, +)$ is an abelian group,
2. $\forall a, b, c \in R$, $a \times (b \times c) = (a \times b) \times c$ (associative law holds), and
3. $\forall a, b, c \in R$, $a \times (b + c) = (a \times b) + (a \times c)$
   and $(b + c) \times a = (b \times a) + (c \times a)$ (distributive law holds).

Note that a ring is commutative if $ab = ba$ $\forall a, b \in R$. Further, R has a unity element if $\exists 1 \in R$ such that $1a = a1$ $\forall a \in R$.

Note that, notationally speaking, the additive identity is typically denoted by 0 and the additive inverse of $a \in R$ is denoted $-a$.

We define subtraction, $a - b$, to be $a - b = a + (-b)$.

Definition 7.1.2: A ring R with unity element 1 that is not equal to 0 is called a division ring if every $a \in R$ with $a \neq 0$ has a multiplicative inverse. A commutative division ring is called a field.

Note that the reason for the term "division ring" is that dividing by a corresponds to multiplying by a^{-1}.

Definition 7.1.3: A ring R that is commutative, has a unity element, and the cancellation property, is called an integral domain. R has the cancellation property if $ab = ac$ with $a \neq 0$ implies $b = c$ and if $ba = ca$ with $a \neq 0$ then $b = c$.

Proposition 7.1.1: Let R be a ring.

1. $0a = a0 = 0, a \in R$,
2. $(-a)b = a(-b) = -(ab)$ $\forall a, b \in R$,
3. $(-a)(-b) = ab$ $\forall a, b \in R$, and
4. if R has unity 1, then $-a = (-1)a$

Proof (1):

$$\begin{aligned} 0a &= (0+0)a \\ &= 0a + 0a \\ \implies 0a &= 0a + 0a \\ \implies 0 &= 0a \text{ by additive cancellation since } (R,+) \text{ is abelian, and} \\ a0 &= a(0+0) \\ &= a0 + a0 \\ \implies a0 &= a0 + a0 \\ \implies 0 &= a0. \ \square \end{aligned}$$

Proof (2): By (1), we have

$$\begin{aligned} 0 = 0b &\implies 0 = (a + (-a))b \\ &\implies 0 = ab + (-a)b \\ &\implies 0 - (ab) = -(ab) + ab + (-a)b \\ &\implies -(ab) = 0 + (-a)b \\ &\implies -(ab) = (-a)b. \ \square \end{aligned}$$

Definition 7.1.4: Let R be a ring. A nonzero element $a \in R$ is called a zero divison if there exists $b \in R$, $b \neq 0$, such that $ab = 0$ or $ba = 0$ if R has unity 1 with $1 \neq 0$.

Note that no ring R can have both zero divisors *and* the cancellation property.

Definition 7.1.5: If a ring R has unity $1 \neq 0$, an element u is called a unit if $\exists v \in R$ such that $uv = 1 = vu$. The units in R are denoted $R^\times$.

Example 7.1.1: In $\mathbb{Z}_6$, $2, 3$, and 4 are zero divisors while 1 and 5 are units.

Example 7.1.2: In $\mathbb{Z}$, there are no zero divisors and the units are 1 and -1.

Example 7.1.3: In $\mathbb{Q}$, all nonzero elements are units.

Definition 7.1.6: If R is a ring with unity 1, then $R^\times$ forms a group under multiplication and is called the group of units.

Example 7.1.4: In $\mathbb{Z}_{10}^\times$, the invertible elements are those relatively prime to 10. That is, $\mathbb{Z}_{10}^\times = \{1, 3, 7, 9\}$.

With this notation, we can say that F is a field if F is commutative and $F^\times = F \backslash \{0\}$.

Note that, if u is a unit in R, then u is not a zero divisor (and visa versa).

Proof: Suppose $ux = 0$ or $xu = 0$ for some $x \in R$. If $ux = 0$, then

$$\begin{aligned} u^{-1}(ux) &= u^{-1}0 \\ (u^{-1}u)x &= 0 \\ 1x &= 0 \\ x &= 0. \end{aligned}$$

If $xu = 0$, then a similar argument shows that $x = 0$.

Proposition 7.1.2: If R is an integral domain (ID) and $ab = 0$, then either $a = 0$ or $b = 0$. Thus, an ID has no zero divisors.

Proof: If $a = 0$ then we are done. Then assume $a \neq 0$. Now from $ab = 0$, we have $ab = a0 \implies b = 0$ (by cancellation of a). $\square$
Note that one can also prove the converse. Further, if R is a commutative ring with unity (cru) and has no zero divisors, then R has the cancellation property. See below for a sketch of the proof:

$$\begin{aligned} ab = ac &\implies ab - ac = 0 \\ &\implies a(b - c) = 0 \\ &\implies a = 0 \text{ or } b - c = 0 \\ &\implies b = c \ \square \end{aligned}$$

Corollary 7.1.1: If R is a finite integral domain, then R is a field.

Proof: Suppose R is a finite integral domain. Then we know R is a commutative ring with unity and no zero divisors. Further, R has the cancellation property. Let $a \in R : a \neq 0$. We need to show that a is a unit. Let $f : R \longrightarrow R$ be the function $f(x) = ax$.

Claim: f is one-to-one. We have

$$\begin{aligned} f(x_1) = f(x_2) &\implies ax_1 = ax_2 \\ &\implies x_1 = x_2 \text{ by cancellation.} \end{aligned}$$

Since $f : R \longrightarrow R$ is one-to-one and R is finite, f is onto. Thus, $\exists b \in R$ such that $f(b) = 1$, so $ab = 1$. Thusly, a is a unit.

Definition 7.1.7: Let R be a ring and $S \subseteq R$. The set S is a subring of R if $(S, +) \leq (R, +)$ and S is closed under multiplication.

The subring test says that S is a subring of R if

1. $S \neq \emptyset$
2. $a, b \in S \Longrightarrow a - b \in S$, and
3. $a, b \in S \Longrightarrow ab \in S$.

Example 7.1.5: Let R be a commutative ring with unity. Let $a \in R$. Show $< a >= \{ar : r \in R\}$ is a subring of R.

Let $ar_1, ar_2 \in < a >$. Then

$$\begin{aligned} ar_1 - ar_2 &= a(r_1 - r_2) \\ &= ar_3, r_3 \in R, \end{aligned}$$

which has the correct form to be an element of $< a >$. Similarly,

$$\begin{aligned} (ar_1)(ar_2) &= a(r1)(ar_2) \\ &= ar_4, r_4 \in R, \end{aligned}$$

which has the correct form to be an element of $< a >$. □

7.2 Examples: Polynomial Rings, Matrix Rings, and Group Rings

Definition 7.2.1: Let R be a commutative ring with unity, and let x be an indeterminant (kind of like a variable). A polynomial in x with coefficients in R is an expression of the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \text{ where } a_i \in R, n \geq 0.$$

This polynomial is monic if $a_n = 1$. The ring of polynomials in x with coefficients in R is denoted $R[x]$. Since R is commutative, so is $R[x]$. Since R has unity (say 1), so does ring $R[x]$ (namely, $u(x) = 1$).

The constant polynomials are those of the form $c(x) = a_0$, where $a_0 \in R$. The set of all constant polynomials forms a subring that is isomorphic to R.

Proposition 7.2.1: Let R be an integral domain and $p(x)$ and $q(x)$ be nonzero polynomials. Then

1. the degree of $p(x)q(x)$ is equal to the degree of $p(x)$ plus the degree of $q(x)$. Symbolically, $\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x))$.
2. the units in $R[x]$ are the units in R (viewed as constant polynomials), and
3. $R[x]$ is an integral domain.

Note that $\deg(c(x)) = 0$, where $c(x)$ is a constant polynomial and $\deg(p(x))$ is undefined if $p(x) = 0$.

Proof (1): Assume $\deg(p(x)) = n$ and $\deg(q(x)) = m$. Let

$$\begin{aligned} p(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, a_n \neq 0 \text{ and} \\ q(x) &= b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0, b_m \neq 0. \end{aligned}$$

Thus $p(x)q(x) = a_n b_m x^{n+m} + \cdots + a_0 b_0$. Further, since integral domains have no zero divisors, $a_n b_m \neq 0$. Then $\deg(p(x)q(x)) = n + m = \deg(p(x)) + \deg(q(x))$. □

Proof (2): Suppose $a(x)$ is a unit in $R[x]$. Then $\exists b(x) \in R[x]$ such that $a(x)b(x) = 1$. Then, by part 1 above, $\deg(a(x))(b(x)) = \deg(1) \Longrightarrow \deg(a(x)) + \deg(b(x)) = 0 \Longrightarrow \deg(a(x)) = \deg(b(x)) = 0$. Then $a(x) = a_0$ and $b(x) = b_0$ for some $a_0, b_0 \in R$. So from $a(x)b(x) = 1$, we get $a_0 b_0 = 1$. Thus a_0 is a unit in R. □

7.3 Ring Homomorphisms and Quotient Rings

Definition 7.3.1: Let R and S be rings. A function $\phi : R \longrightarrow S$ is a ring homomorphism if

$$\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2) \text{ and}$$
$$\phi(r_1 r_2) = \phi(r_1)\phi(r_2) \ \forall r_1, r_2 \in R.$$

A ring homomorphism that is one-to-one is a ring isomorphism.

Definition 7.3.2: The kernel of a ring homomorphism is

$$\ker(\phi) = \{r \in R : \phi(r) = 0_S\}.$$

Proposition 7.3.1: Let $\phi : R \longrightarrow S$ be a ring homomorphism.

1. The image of ϕ, which is $Im(\phi) = \{s \in S : \exists r \in R \text{ with } \phi(r) = s\}$ is a subring of S.
2. $\ker(\phi)$ is a subring of R.

Definition 7.3.3: Let R be a ring and I be a subset of R. Define

$$rI = \{ra : a \in I\} \text{ and } Ir = \{ar : a \in I\}.$$

We say that I is a left ideal if I is a subring and $rI \subseteq I \ \forall r \in R$. We say that I is a right ideal if I is a subring and $Ir \subseteq I \ \forall r \in R$. I is a two-sided ideal, also just called an ideal, if I is both a left and a right ideal.

Since R is an abelian group, we have additive cosets. If I is a subring of R, let

$$r + I = \{r + a : a \in I\}.$$

Then $r + I$ is a coset of I. Note that we write R/I for the set of all cosets of I in R.

Proposition 7.3.2: Let R be a ring and I be an ideal of R (denoted $I \trianglelefteq R$). The additive quotient group R/I is a ring under the operations

$$(r + I) + (s + I) = r + s + I \tag{1}$$
$$(r + I)(s + I) = rs + I. \tag{2}$$

Proof (2): Our goal is to prove that $(r + I)(s + I) = rs + I$ is well-defined under the hyptohesis that I is an ideal. Suppose that $r_1 + I = r_2 + I$ with r_1 not necessarily equal to r_2 and $s_1 + I = s_2 + I$ with s_1 not necessarily equal to s_2. We need to show $r_1 s_1 + I = r_2 s_2 + I$. *Note that what we need to show is equivalent to* $r_1 s_1 - r_2 s_2 \in I$.

Now, $r_1 + I = r_2 + I$ tells us $r_2 - r_1 \in I$ and $s_1 + I = s_2 + I$ tells us $s_2 - s_1 \in I$. Then

$$\begin{aligned}(r_2 - r_1)(s_2 - s_1) \in I &\Longrightarrow r_2 s_2 - r_2 s_1 - r_1 s_2 + r_1 s_1 \in I \\ &\Longrightarrow r_1(s_1 - s_2) + (r_2 - r_1)s_2 \in I.\end{aligned}$$

This element, and all components of it, are in I, and $r_1(s_1 - s_2) + (r_2 - r_1)s_2 = r_1 s_1 - r_2 s_2$. □

Definition 7.3.4: If $I \trianglelefteq R$, we call the ring R/I a quotient ring. That is, if I is an ideal of R, we call the set of all cosets of I in R a quotient ring.

Theorem 7.3.1: The First Isomorphism Theorem for Rings

1. If $\phi : R \longrightarrow S$ is a ring homomorphism, then $\ker(\phi) \trianglelefteq R$, $Im(\phi)$ is a subring of S, and $R/\ker(\phi) \cong Im(\phi)$.

2. If $I \trianglelefteq R$, then $\psi : R \longrightarrow {}^{R}/_{I}$ defined by $\psi(r) = r + I$ is an onto ring homomorphism with kernel I. This map is the natural projection map, or canonical quotient map.

Let I and J be ideals of R. Then

1. the sum of I and J is $I + J = \{i + j : i \in I, j \in J\}$,
2. the product of I and J, denoted IJ, is all expressions of the form $a_m b_m + a_{m-1} b_{m-1} + \cdots + a_1 b_1, m > 0, a_i \in I, b_i \in J$, and
3. for $n \geq 1$, the nth power of I, denoted I^n, is all finite sums whose terms are of the form $a_1 a_2 \cdots a_n$ where $a_i \in I$.

Example 7.3.1: In $\mathbb{Z}$, let $I = 6\mathbb{Z} = \{6m : m \in \mathbb{Z}\}$ and $J = 10\mathbb{Z} = \{10n : n \in \mathbb{Z}\}$. Then $I + J = 2\mathbb{Z}$. Note that, in general, if you take $m\mathbb{Z} + n\mathbb{Z}$, you get $d\mathbb{Z}$, where $d = \gcd(m, n)$.

Theorem 7.3.2: Let R be a ring. Then

1. the second isomorphism theorem for rings says if A is a subring of R and $B \trianglelefteq R$, then $A + B$ is a subring, $B \trianglelefteq A + B$, and $A \cap B \trianglelefteq A$. Then
$$ {}^{A+B}/_{B} \cong {}^{A}/_{A \cap B}, \text{ and} $$
2. the third isomorphism theorem for rings says that if $I, J \trianglelefteq R$ with $I \subseteq J$, then ${}^{J}/_{I} \trianglelefteq {}^{R}/_{I}$ and
$$ {}^{R/_{I}}/_{J/_{I}} \cong {}^{R}/_{J}. $$

7.4 Properties of Ideals

Definition 7.4.1: Let $A \subseteq R$ where R is a ring. Let (A) denote the smallest ideal in R that contains A. Then (A) is the ideal generated by A and
$$ (A) = \bigcap_{I \text{ is an ideal with } A \subseteq I} I. $$

When $A = \{a\}$ we write (a) and say (a) is the principal ideal generated by a.

Now, assume R is a commutative ring with unity. In this case, we can prove that $I = \{ar : r \in R\}$ is an ideal in R that contains a. Note that $a \cdot 1 = a \in I$, so $I \neq \emptyset$. Then, let $ar_1, ar_2 \in I$ and $s \in R$. Then
$$ \begin{aligned} ar_1 - ar_2 &= a(r_1 - r_2) \\ &= ar_3, r_3 \in R \Longrightarrow ar_1 - ar_2 \in I. \end{aligned} $$

Also, we have
$$ \begin{aligned} s(ar_1) &= a(sr_1) \in I \text{ and} \\ (ar_1)s &= a(r_1 s) \in I. \end{aligned} $$

Then $I \trianglelefteq R$ and $a \in I$. By definition of (a), $(a) \subseteq I$. Now, if $ar \in I$, then, since (a) is an ideal that contains a, it also contains ar by the absorbs products property, so $I \subseteq (a)$. So, in a commutative ring with unity, $(a) = \{ar : r \in R\}$. $\square$

In this case, we have that $b \in (a)$ if and only if $b = ar$, $r \in R$. Here, we would say $a|b$ and b is a multiple of a.

Proposition 7.4.1: Let $I \trianglelefteq R$ with R having a unity element. Then

1. $I = R \Longleftrightarrow I$ contains a unit, and
2. if R is commutative, then R is a field if and only if the only ideals in R are $\{0\}$ and R.

Proof (1): First, assume $I = R$. Then $1_R \in I$ and 1_R is a unit. For the other direction, suppose $u \in I$ with u a unit, say $vu = 1_R$. Then, by the absorbs products property, $vu = 1_R$ is in I since $u \in I$. Therefore, for any $r \in R$, $r1_R \in I$, so $I = R$. □

Proof (2): Assume R is a field and $J \trianglelefteq R$. If $J = \{0\}$, then we are done, so assume $J \neq \{0\}$. Then J contains a nonzero element u. Since R is a field, u is a unit. So by part 1, $J = R$.

For the other direction, assume $\{0\}$ and R are the only ideals. Let $r \in R$ with $r \neq 0$. We want to show r is a unit. Now $\{0\} \neq (r)$, so $(r) = R$. But $1_R \in R$, so $1_r \in (r)$. Thus, $\exists b \in R : 1_R = br$ and thus r is a unit. □

Corollary 7.4.1: If R is a field and $\phi : R \longrightarrow S$ is a nontrivial ring homomorphism, then ϕ is one-to-one.

Proof: We know $\ker(\phi) \trianglelefteq R \implies \ker(\phi) = \{0\}$ or $\ker(\phi) = R$ since R is a field. Since ϕ is nontrivial, $\ker(\phi) \neq R$. Then $\ker(\phi) = \{0\}$, so ϕ is one-to-one. □

Definition 7.4.2: Let S be a ring and $M \trianglelefteq S$ with $M \neq S$. We call M a maximal ideal if whenever $I \trianglelefteq S$ with $M \subseteq I \subseteq S$, we have either $M = I$ or $M = S$.

Proposition 7.4.2: Let R be a commutative ring with unity with $M \trianglelefteq R$. Then R/M is a field if and only if M is maximal.

Proof: Assume M is a maximal ideal. Since R is a commutative ring with unity, so is R/M. Now we show every $r + M \neq 0 + M$ in R/M is invertible. Consider $I = \{m + sr : m \in M, s \in R\}$. Then I is an ideal. Certainly, $M \subseteq I$ and $R \subseteq I$, but M is maximal, so $I = R$.

Therefore, $\exists m_1 \in M$ and $\exists s_1 \in R$ such that $1_R = m_1 + s_1 r$. Now, $s_1 r - 1_R = m$ so $s_1 r - 1_R \in M$. Then this implies

$$\begin{aligned} s_1 r + M = 1_R + M &\implies (s_1 + M)(r + M) = 1_R + M \\ &\implies r + M \text{ is invertible.} \end{aligned}$$

For the other direction, assume R/M is a field. Let I be an ideal with $M \subseteq I \subseteq R$ where $M \neq I$. We will now argue that $I = R$. Let $r \in I$ with $r \notin M$. Since R/M is a field and we have $r + M \neq 0 + M$, the element $r + M$ has an inverse. Let $(r + M)(s + M) = 1_R + M$. Then

$$\begin{aligned} sr + M = 1_R + M &\implies 1_R - sr = m,\ m \in M \\ &\implies 1_R = sr + m. \end{aligned}$$

Now, since $r \in I$ and I absorbs products, $sr \in I$. Then $1_R \in I$ and $I = R$. Now, if an ideal contains a unit, it must be the whole ring. This shows that M is maximal. □

Proposition 7.4.3: Assume R is a commutative ring with unity. Then P is a prime ideal if and only if R/P is an integral domain.

Proof: Since R is a commutative ring with unity, R/P is also a commutative ring with unity. Then we need only show R/P has no zero divisors. Suppose $(a + P)(b + P) = 0 + P$. Then $(a + P)(b + P) = ab + P$ and $ab \in P$, so we must have at least one of $a \in P$ or $b \in P$. Then $a + P = 0 + P$ or $b + p = 0 + P$.

For the other direction, assume R/P is an integral domain, and let $ab \in P$. Then

$$\begin{aligned} ab \in P &\implies ab + P = 0 + P \\ &\implies (a + P)(b + P) = 0 + P \\ &\implies a + P = 0 + P \text{ or } b + P = 0 + P \\ &\implies a \in P \text{ or } b \in P. \ \square \end{aligned}$$

Note that, if M is a maximal ideal, then M is prime. Further, note that the converse of this statement is not necessarily true.

Example 7.4.1: In $\mathbb{Z}$, the (0) ideal is prime but not maximal. Further, (x) is prime in $\mathbb{Z}[x]$ but not maximal. Also note that $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, which is an integral domain but not a field.

7.5 Rings of Fractions

In this section, assume R is an integral domain. Our goal is to construct a new ring that

1. contains R,
2. is a field, and
3. is minimal with respect to (1) and (2).

Given R, let $S = \{(a, b) : a, b \in R, b \neq 0\}$. Define an equivalence relation $\sim$ on S by the rule $(a, b) \sim (c, d)$ if and only if $ad = bc$. Let $Frac(R)$ be the set of all equivalence classes under the relation $\sim$ on S. Given $[a, b], [c, d] \in Frac(R)$, define

$$[a, b] + [c, d] = [ad + bc, bd] \text{ and}$$
$$[a, b][c, d] = [ac, bd].$$

With these two operations, $Frac(R)$ is a field. The additive identity is $[0, 1_R]$, and the multiplicative identity is $[1_R, 1_R]$. Given $[a, b]$ where $[a, b] \neq [0, 1_R]$, observe $[a, b][b, a] = [ab, ba] = [1_R, 1_R]$. That is, any nonzero element is invertible.

The set $\{[a, 1_R] : a \in R\}$ is a subset of $Frac(R)$ that is isomorphic ro R. This field is called the field of fractions of R.

Some good facts to know:

1. If R is a field, then $Frac(R) \cong R$.
2. If $R = \mathbb{Z}$, then $Frac(R) = \mathbb{Q}$.
3. If R is an integral domain, then $R[x]$ is an integral domain and $Frac(R[x]) = R(x)$, where $R(x)$ is the set of rational functions. That is, where $R(x) = \left\{ \dfrac{f(x)}{g(x)} : f(x), g(x) \in R[x], g(x) \neq 0 \right\}$.